



Robust Multi-label Classification via Preference Learning

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Abstract

In this paper, we explore how multi-label classification (MLC) tasks can be cast into order structure learning. Our motivation for doing so is to exploit the very rich structure of the orders to improve and robustify MLC learning. We describe formally how MLC can be transformed into an order structure learning and prediction task, and then proceed to study the problem of predicting Bayes-optimal order structures. We then perform some experiments in settings where the use of order structures can be very beneficial: robust MLC in the presence of noisy and imbalanced labels, and making MLC predictions with partial abstention.

Keywords MLC · Preference learning · Noisy and imbalanced labels · Robustness

Vu-Linh Nguyen and Xuan-Truong Hoang have contributed equally to this work and should be regarded as co-first authors.

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1 Introduction

In multi-label classification (MLC), each instance can be associated with zero, one, or several labels simultaneously (see (Bogatinovski et al., 2022; Zhang & Zhou, 2014) and references therein). Probabilistic MLC (Nguyen et al., 2023) consists of learning a probabilistic classifier, for instance by maximizing a conditional likelihood, that produces a joint probability over the possible label subsets. Such a probability provides useful information about the prediction uncertainty, and also allows the derivation of a Bayes-optimal prediction (BOP) that minimizes a given expected loss/accuracy (Dembczyński et al., 2012; Nguyen & Hüllermeier, 2021; Waegeman et al., 2014).

In this paper, we propose to recast the MLC task into a preference learning problem (Fürnkranz et al., 2008; Masson et al., 2016), where we allow the predictor to output preorders, i.e., relations that are transitive and irreflexive, and for which two elements can be incomparable or indifferent. The use of order structures to solve the MLC task is not new, with the calibrated label ranking (CLR) coming first to mind. However, CLR considers complete orders, and we discuss in details the limitations of considering such a sub-family in Appendix B. We are however not aware, within probabilistic settings, of methods aiming at learning optimal preorders to solve the MLC task. This contrasts with the label ranking tasks, where such procedures have been proposed in the past (Cheng et al., 2012), as well as with MLC solutions using imprecise probabilistic models (Masson et al., 2016; Moral-García & Destercke, 2022; Moral-García et al., 2022). As those approaches use different settings than probabilistic MLC, we will not discuss them in details, but invite the interested reader to consult provided references.

After providing a minimal description of probabilistic MLC as well as how MLC samples can be seen as observations of specific orders between labels in Sect. 2, we present in Sect. 3 a formal framework for learning Bayes-optimal orders (BOOs) in MLC. Under this setting, we predict, for each instance, an order within a pre-defined family, instead of a set of relevant labels. We shall discuss different notions of order structures, including preorders and partial orders. Their representation capacity is discussed and compared to related types of order structures in Appendix B. Section 4 then deals with the computational aspects of finding BOOs. Section 5 presents two sets of experiments to assess the usefulness of the proposed framework: making robust predictions in the presence of noisy and imbalanced labels (Tarekegn et al., 2021), and making predictions with partial abstention (Nguyen & Hüllermeier, 2021). Finally, Sect. 6 concludes this work and sketches out potential follow-up works. Our complete training pipeline is summarised by Fig. 1, with reference to the sections or appendices that deal with each problem.

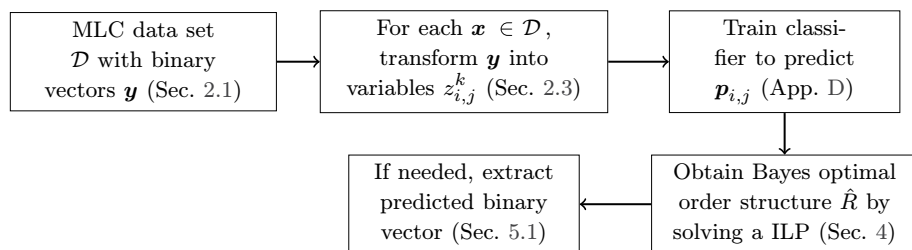


Fig. 1 Training pipeline summary

2 Multi-label Classification and Orders

This section introduces notations on MLC as well as its connection with order structures. It also establishes the prediction problem treated in this paper.

2.1 Probabilistic Multi-label Classification

Let $\mathcal{X}$ denote an instance space, and let $\mathcal{L} = \{\lambda_1, \dots, \lambda_K\}$ be a finite set of labels. We assume that an instance $x \in \mathcal{X}$ is (probabilistically) associated with a subset of relevant labels $\Lambda = \Lambda(x) \in 2^{\mathcal{L}}$ and the complement $\mathcal{L} \setminus \Lambda$ is considered as irrelevant for x . We identify a set Λ of relevant labels with a binary vector $\mathbf{y} = (y^1, \dots, y^K)$, where $y^k = \llbracket \lambda_k \in \Lambda \rrbracket$.¹ By $\mathcal{Y} = \{0, 1\}^K$ we denote the set of possible labeling. Some frequently used notations and acronyms are listed in the Table 6 of Appendix A.

We assume that an instance $x \in \mathcal{X}$ is (probabilistically) associated with members of $\mathcal{Y}$. We denote by $p(Y|x)$ the joint conditional probability distribution of Y given $X = x$. Given training data $\mathcal{D} = \{(x_n, \mathbf{y}_n) | n = 1, \dots, N\}$, which is drawn independently from $p(X, Y)$, probabilistic MLC will aim at estimating a conditional probability $p(\mathbf{y}|x)$, for instance by optimizing a regularized conditional log-likelihood (Murphy, 2012; Nguyen et al., 2023), and then to derive BOPs from such an estimate. In this paper, we will be mainly interested in this latter aspect, assuming the estimation of $\hat{p}(\mathbf{y}|x)$, $\mathbf{y} \in \mathcal{Y}$, has been performed.

In this case, and given a metric $f : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ such that a higher metric gives better results, the BOP of an instance x is defined as

$$\hat{\mathbf{y}} = \hat{\mathbf{y}}(x) \in \operatorname{argmax}_{\bar{\mathbf{y}} \in \mathcal{Y}} E_{\mathbf{y}}(f(\mathbf{y}, \bar{\mathbf{y}})) = \operatorname{argmax}_{\bar{\mathbf{y}} \in \mathcal{Y}} \sum_{\mathbf{y} \in \mathcal{Y}} f(\mathbf{y}, \bar{\mathbf{y}}) p(\mathbf{y}|x), \quad (1)$$

with different MLC metrics resulting in different predictions (Dembczyński et al., 2012; Waegeman et al., 2014). In general, knowing the conditional probability distribution $p(Y|x)$ is necessary for finding the prediction for any metric f , with some such as Hamming loss only requiring to know some values of the distribution (Dembczyński et al., 2012).

However, it is quite common to consider, within MLC problems, predictions that have a different nature than binary vectors $\{0, 1\}^K$, for instance, rankings (Dembczyński et al., 2012; Fürnkranz & Hüllermeier, 2010; Hüllermeier et al., 2008; Nguyen & Hüllermeier, 2021) where one aims to order labels from the most relevant to the least relevant, or partial vectors where one considers the output space $\mathbf{y} \in \mathcal{Y}_{\text{abs}} := \{0, 1, \mathbb{R}\}^K$, where $y^k = \mathbb{R}$ indicates that the classifier abstains from making a precise prediction on the label k -th (Nguyen & Hüllermeier, 2021). It is then desirable to design new metrics that adapt to such prediction spaces.

In this paper, we are interested in considering not complete rankings (Fürnkranz & Hüllermeier, 2010) nor partial vectors (Nguyen & Hüllermeier, 2021), but preorders and partial orders, which are more general representations that, in particular, encompass the two other cases we just mentioned.

¹ $\llbracket \cdot \rrbracket$ is the indicator function, i.e., $\llbracket A \rrbracket = 1$ if the predicate A is true and $= 0$ otherwise.

2.2 Reminder on Orders

Let us first remind some aspects of preorders and partial orders, the two structures that we will consider, and the notations we will associate with them. A *preorder* is a binary relation $R \subseteq \mathcal{L} \times \mathcal{L}$ that is reflexive and transitive, that can encode strict preferences, incomparability, and indifference, and for which we will use the following notation:

$$\begin{aligned} \lambda_i \succ_R \lambda_j \quad (\text{or simply } \lambda_i \succ \lambda_j) & \quad \text{if } r_{i,j} = 1, r_{j,i} = 0, \\ \lambda_i \perp_R \lambda_j \quad (\text{or simply } \lambda_i \perp \lambda_j) & \quad \text{if } r_{i,j} = 0, r_{j,i} = 0, \\ \lambda_i \sim_R \lambda_j \quad (\text{or simply } \lambda_i \sim \lambda_j) & \quad \text{if } r_{i,j} = 1, r_{j,i} = 1, \end{aligned} \quad (2)$$

where $r_{i,j} = 1$ if $(\lambda_i \succ_R \lambda_j) \in R$ and $r_{i,j} = 0$ if $(\lambda_i \succ_R \lambda_j) \notin R$. To be a valid preorder, R should satisfy the transitivity property

$$r_{i,k} + r_{k,j} - 1 \leq r_{i,j}, \forall i \neq j \neq k. \quad (3)$$

We shall encode each preorder R , by the corresponding set of pairwise preference relations, i.e., $R := \{z_{i,j}^l \mid 1 \leq i < j \leq K, 1 \leq l \leq 4\}$, where

$$\begin{aligned} z_{i,j}^1 &= \llbracket \lambda_i \succ_R \lambda_j \rrbracket = r_{i,j}(1 - r_{j,i}), & z_{i,j}^2 &= \llbracket \lambda_j \succ_R \lambda_i \rrbracket = (1 - r_{i,j})r_{j,i}, \\ z_{i,j}^3 &= \llbracket \lambda_i \perp_R \lambda_j \rrbracket = (1 - r_{i,j})(1 - r_{j,i}), & z_{i,j}^4 &= \llbracket \lambda_i \sim_R \lambda_j \rrbracket = r_{i,j}r_{j,i}. \end{aligned} \quad (4)$$

Note that only one of those values can be equal to one. A partial order, on the other hand, is a structure that is reflexive, antisymmetric, and transitive, and that can encode strict preferences and incomparability. It therefore forbids the configuration $r_{i,j} = 1, r_{j,i} = 1$ unless $i = j$. Note that other commonly considered structures such as complete preorders (forbidding $r_{i,j} = 0, r_{j,i} = 0$) or complete orders (only allowing $r_{i,j} \neq r_{j,i}$) have less flexibility, and for the later have already been extensively explored within the MLC literature (Nguyen & Hüllermeier, 2021). We will denote by $\mathcal{R}_{\text{pre}}$ and $\mathcal{R}_{\text{par}}$ the sets of possible preorders and partial orders over $\mathcal{Y}$.

2.3 From Binary Vectors to Order Structures

In order to cast multi-label learning into order learning problems, we need to map binary vectors of relevant labels into order structures. We can translate each possible labeling $\mathbf{y} \in \mathcal{Y}$ into a preorder R using an injective non-surjective map $T_{\text{pre}} : \mathcal{Y} \longrightarrow \mathcal{R}_{\text{pre}}$, where $R = T_{\text{pre}}(\mathbf{y})$ is constructed by using the binary relations

$$\begin{aligned} r_{i,j} &= 1, r_{j,i} = 0 & \text{if } y^i = 1, y^j = 0, \\ r_{i,j} &= 0, r_{j,i} = 1 & \text{if } y^i = 0, y^j = 1, \\ r_{i,j} &= 0, r_{j,i} = 0 & \text{if } y^i = 0, y^j = 0, \\ r_{i,j} &= 1, r_{j,i} = 1 & \text{if } y^i = 1, y^j = 1, \end{aligned} \quad (5)$$

Similarly, we will adopt the following mapping $T_{\text{par}} : \mathcal{Y} \longrightarrow \mathcal{R}_{\text{par}}$, where $R = T_{\text{par}}(\mathbf{y})$ for partial orders

$$\begin{aligned}
 r_{i,j} &= 1, r_{j,i} = 0 & \text{if } y^i = 1, y^j = 0, \\
 r_{i,j} &= 0, r_{j,i} = 1 & \text{if } y^i = 0, y^j = 1, \\
 r_{i,j} &= 0, r_{j,i} = 0 & \text{if } y^i + y^j \neq 1.
 \end{aligned} \tag{6}$$

The first encoding assumes that while the set of relevant labels are indifferent between them, i.e., they are all equally relevant, the set of irrelevant labels cannot really be compared with each other, as we lack information to compare them. Incomparabilities can also be used to encode missing information about the relevance of some labels. The second encoding considers that we just have insufficient information to compare relevant and irrelevant labels between them. Again, we do not consider the choice where all relevant and irrelevant labels are indifferent between themselves, as this would be very close to our second encoding, while constraining more the resulting predictions (the indifference relation having to be an equivalence relation, which is not the case of the incomparability relation).

Example 1 Consider the case where we have four labels $\lambda_1, \dots, \lambda_4$ and where the observed output for x is $y = (1, 1, 0, 0)$. The resulting orders are pictured in Fig. 2. The partial order variables are

$$\begin{aligned}
 z_{1,2}^4 &= 1, z_{1,3}^1 = 1, z_{1,4}^1 = 1; z_{2,1}^4 = 1, z_{2,3}^1 = 1, z_{2,4}^1 = 1 \\
 z_{3,1}^2 &= 1, z_{3,2}^2 = 1, z_{3,4}^3 = 1; z_{4,1}^2 = 1, z_{4,2}^2 = 1, z_{3,4}^3 = 1.
 \end{aligned}$$

which will then be used as observation to learn a model that will then predicts probabilities of variables $z_{i,j}^k$ of future observations.

We can already see in Fig. 2 that using order structure offers a richer language than the ones of binary vectors, as the same binary vector can be mapped into different structures. This richness offer a large playground, possibly allowing one to account between conflicting information about an instance, for example when training examples nearby this instance contain conflicting information, e.g., sometimes considering that λ_i and λ_j are relevant, and sometimes indicating that only λ_i is relevant. One can think, for instance, of images that are very similar, but such that a given object is present in one and absent in the other.

3 Predicting Bayes-Optimal Orders

Having set out the setting, we now deal with the problem of learning and predicting optimal order structures, in particular when initial observations are of the multi-label kind.

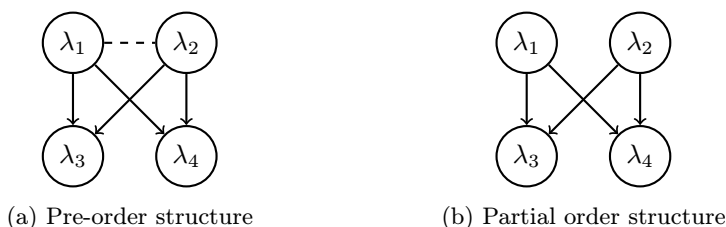


Fig. 2 From multi-label output to order structures

3.1 Probabilistic Orders

Switching to the problem of predicting order structures, we now consider that to each query instance x is associated a conditional joint probability distribution over the possible order structures $\mathcal{R}$, denoted by $p(\mathcal{R} | x) := \{p(R | x) | R \in \mathcal{R}\}$. However, from now on and to avoid too many notations, we will drop the conditioning, and will consider that our probabilities are conditional upon an observation x .

Given a metric $f : \mathcal{R} \times \mathcal{R} \rightarrow \mathbb{R}_+$, which compares a prediction $\hat{R} \in \mathcal{R}$ with a ground-truth $R \in \mathcal{R}$, a BOO of any “the higher the better” metric f is defined as

$$\hat{R} = \hat{R}(x) \in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} E_R(f(R, \bar{R})) = \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{R \in \mathcal{R}} f(R, \bar{R}) p(R). \quad (7)$$

It is clear that the above formulation may quickly turn out to be intractable, as the spaces $\mathcal{R}$ are combinatorial in nature. This section, therefore, explores how we can make this problem tractable in some cases and how classical MLC metrics can be cast in this setting. Note also that Eq. (7) can be used for any problems where predicting order structures makes sense, and its use goes beyond MLC. For example, solving (7) is equivalent to solving the problem of finding a center/median of a given set of order structures in the clustering setting (Viapiani, 2015) and preference aggregation (Barthelemy et al., 1989). Therefore, what has/will been/be done in our setting would benefit the development of these two relevant research directions and vice versa.

3.2 Metrics

In this paper, we adopt a straightforward way to adapt various MLC metrics f_{MLC} (Nguyen & Hüllermeier, 2021; Powers, 2011; Zhang & Zhou, 2014), such as Hamming accuracy, subset 0/1 accuracy, F-measure, and so on, to the case of order structures. For this, it is sufficient to notice that an order can be summarized by a 0-1 vector bearing on $z_{i,j}^l$ values given by Eq. (4), reminding that $z_{i,j}^l$ values (4) are defined for preorders and partial orders using $r_{i,j}$ defined in (5) and (6), respectively.

Given the transformations $T \in \{T_{\text{pre}}, T_{\text{par}}\}$ and $R := T(y)$, we propose to define its corresponding metric f_{OS} considering order structures as follows:

$$f_{\text{OS}}(y, \hat{R}) := f_{\text{MLC}}(T(y), \hat{R}) = f_{\text{MLC}}(R, \hat{R}). \quad (8)$$

By definition (8), the Hamming accuracy (9) and subset 0/1 accuracy (10)

$$f_{\text{MLC}}^{\text{ham}}(y, \hat{y}) = \sum_{k=1}^K \mathbb{I}[y^k = \hat{y}^k], \quad (9)$$

$$f_{\text{MLC}}^{\text{sub}}(y, \hat{y}) = \mathbb{I}[y = \hat{y}] \quad (10)$$

can be respectively generalized as $f_{\text{OS}}^{\text{ham}}$ and $f_{\text{OS}}^{\text{sub}}$:

$$f_{\text{OS}}^{\text{ham}}(R, \hat{R}) = \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \llbracket z_{i,j}^l = \hat{z}_{i,j}^l \rrbracket z_{i,j}^l, \quad (11)$$

$$f_{\text{OS}}^{\text{sub}}(R, \hat{R}) = \llbracket R = \hat{R} \rrbracket. \quad (12)$$

where $L = 4$ and 3 for preorders and partial orders, respectively.

Other metrics, including the use of F -measure and Jaccard index (Waegeman et al., 2014), are discussed in details in Appendix E.1. Moreover, one can, in principle, implement our proposal with other metrics assessing the (dis)similarity between pairs of order structures, such as the one discussed in (Yiru et al., 2021). For example, one might also think of adopting cost-sensitive Hamming accuracies, which also provide chances to incorporate the way the end-user wishes to assess the predicted order structures in specific applications via the choice of weights, as discussed in Appendix E.1.

Example 2 Let us consider again the case of four labels, with Fig. 3 picturing an observation transformed into a corresponding order, together with its prediction. We do have

$$\begin{aligned} z_{1,2}^4 = 1, z_{1,3}^1 = 1, z_{1,4}^1 = 1; z_{2,1}^4 = 1, z_{2,3}^1 = 1, z_{2,4}^1 = 1 \\ z_{3,1}^2 = 1, z_{3,2}^1 = 1, z_{3,4}^3 = 1; z_{4,1}^2 = 1, z_{4,2}^1 = 1, z_{3,4}^3 = 1. \end{aligned}$$

and

$$\begin{aligned} \hat{z}_{1,2}^3 = 1, \hat{z}_{1,3}^1 = 1, \hat{z}_{1,4}^1 = 1; \hat{z}_{2,1}^3 = 1, \hat{z}_{2,3}^1 = 1, \hat{z}_{2,4}^3 = 1 \\ \hat{z}_{3,1}^2 = 1, \hat{z}_{3,2}^1 = 1, \hat{z}_{3,4}^3 = 1; \hat{z}_{4,1}^2 = 1, \hat{z}_{4,2}^1 = 1, \hat{z}_{3,4}^3 = 1. \end{aligned}$$

for which we do have $f_{\text{OS}}^{\text{ham}}(R, \hat{R}) = 4$ and $f_{\text{OS}}^{\text{sub}}(R, \hat{R}) = 0$.

From this example, we can already see that $f_{\text{OS}}^{\text{sub}}$ will be much less discriminative, as only one mistake will make it null, while $f_{\text{OS}}^{\text{ham}}$ will make a difference between an order structure close to the ideal one, and one that is the opposite of the ideal one.

3.3 Bayes-Optimal Orders

Let us now deal with how to find optimal order structures for the 0/1 subset and Hamming accuracies, whose weighted variants (E10) are discussed in Appendix E.1.

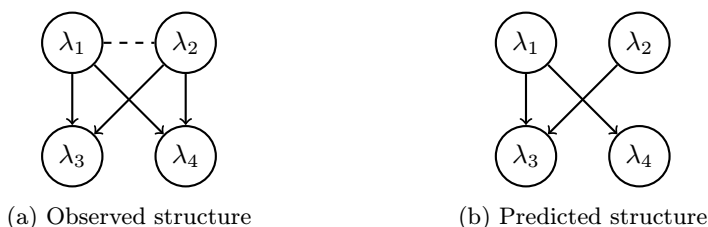


Fig. 3 Metric illustration

3.3.1 General Formulation

Let us first consider the subset 0/1 accuracy. We do have the following rather direct result.

Proposition 1 Let $\mathcal{R} \in \{\mathcal{R}_{\text{pre}}, \mathcal{R}_{\text{par}}\}$. For each query instance x , a BOO (7) of the subset 0/1 accuracy (12) is an order structure $\hat{R} \in \mathcal{R}$ with the highest conditional probability, i.e.,

$$\hat{R} \in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} p(\bar{R}). \quad (13)$$

A similar result can be formulated for the Hamming extension to order structures.

Proposition 2 Let $\mathcal{R} \in \{\mathcal{R}_{\text{pre}}, \mathcal{R}_{\text{par}}\}$. For each query instance x , a BOO (7) of the Hamming accuracy (11) is an order structure $\hat{R} \in \mathcal{R}$ with the highest expected Hamming accuracy, i.e.,

$$\hat{R} \in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L z_{i,j}^l p(\bar{z}_{i,j}^l), \quad (14)$$

with $L = 4$ and 3 for $\mathcal{R}_{\text{pre}}$ and $\mathcal{R}_{\text{par}}$, respectively.

Note that in practice, the only thing that changes is the space over which $\bar{R}$ is sought, or, otherwise said, the space over which we will search for the optimal prediction. This shows the generality of the approach, while emphasizing possible differences in prediction quality when casting them into MLC predictions, or when considering computational aspects. We deal with this aspect in the next section.

3.3.2 Independence Assumption

In general, obtaining the distribution $p(R)$ without simplifying assumptions is neither tractable nor desirable, as the size $|\mathcal{R}|$ has at least a factorial growth with the number of labels. A common assumption is then to assume that the distribution $p(R)$ can be factorized into the probability of pairwise relations.

If we make a simplifying assumption of independence on the pairwise preference relations (Masson et al., 2016; Nguyen et al., 2018), for any $R \in \mathcal{R}_{\text{pre}}$, we have

$$p(R) = \prod_{\lambda_i \succ_R \lambda_j} p(\lambda_i \succ_R \lambda_j) \prod_{\lambda_j \succ_R \lambda_i} p(\lambda_j \succ_R \lambda_i) \prod_{\lambda_j \perp_R \lambda_i} p(\lambda_j \perp_R \lambda_i) \prod_{\lambda_j \sim_R \lambda_i} p(\lambda_j \sim_R \lambda_i), \quad (15)$$

where, for any pair (λ_i, λ_j) , the conditional probabilities on pairwise preference relations can be defined as the conditional probabilities on the corresponding events:

$$p(\lambda_i \succ_R \lambda_j) := p_{i,j}^{1,0} := p(y_i = 1, y_j = 0), \quad (16)$$

$$p(\lambda_j \succ_R \lambda_i) := p_{i,j}^{0,1} := p(y_i = 0, y_j = 1), \quad (17)$$

$$p(\lambda_j \perp_R \lambda_i) := p_{i,j}^{0,0} := p(y_i = 0, y_j = 0), \quad (18)$$

$$p(\lambda_j \sim_R \lambda_i) := p_{i,j}^{1,1} := p(y_i = 1, y_j = 1). \quad (19)$$

Given this factorization, Eq. (13) becomes

$$\begin{aligned} \hat{R} \in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} & \prod_{\lambda_i \succ_{\bar{R}} \lambda_j} p(\lambda_i \succ_{\bar{R}} \lambda_j) \prod_{\lambda_j \succ_{\bar{R}} \lambda_i} p(\lambda_j \succ_{\bar{R}} \lambda_i) \\ & \prod_{\lambda_j \perp_{\bar{R}} \lambda_i} p(\lambda_j \perp_{\bar{R}} \lambda_i) \prod_{\lambda_j \sim_{\bar{R}} \lambda_i} p(\lambda_j \sim_{\bar{R}} \lambda_i). \end{aligned} \quad (20)$$

Note that just optimizing (20) in a pairwise way is straightforward, but will generally not produce a preorder structure. Also, while in the conventional MLC setting, an optimal prediction (1) of the subset 0/1 accuracy (10) under the assumption of label independence is also an optimal prediction (1) of the Hamming accuracy (9) (Nguyen & Hüllermeier, 2021), this will not generally be the case here, indicating a clear difference between the two approaches. These two facts are shown by the next example.

Example 3 Assume the conditional probabilities (16–19) are given in Table 1.

The collection of pairwise preference relations that has the highest probability per (15) is $\{y_1 \perp y_2, y_1 \prec y_3, y_2 \sim y_3\}$, with $0.6 \cdot 0.7 \cdot 0.7 = 0.294$. It also has the highest expected Hamming accuracy (14), i.e., $0.6 + 0.7 + 0.7 = 2$. However, it does not form a preorder because the transitivity (3) is not satisfied. For example, $r_{1,3} + r_{3,2} - 1 = 1 + 1 - 1 = 1 > r_{1,2} = 0$.

One can verify that the only BOO (20) of the subset 0/1 accuracy (12) under the assumption of label independence is $\{y_1 \perp y_2, y_1 \prec y_3, y_2 \perp y_3\}$ with probability (15) equals $0.6 \cdot 0.7 \cdot 0.3 = 0.126$. $\{y_1 \prec y_2, y_1 \prec y_3, y_2 \sim y_3\}$ is the second most probable preorder with probability (15) $0.25 \cdot 0.7 \cdot 0.7 = 0.1225$. However, the only BOO (14) for the Hamming accuracy (11) is $\{y_1 \prec y_2, y_1 \prec y_3, y_2 \sim y_3\}$ with the expected Hamming accuracy (14) equals $0.25 + 0.7 + 0.7 = 1.65$. $\{y_1 \perp y_2, y_1 \prec y_3, y_2 \perp y_3\}$ is the second most probable preorder with the expected Hamming accuracy (14) equals $0.6 + 0.7 + 0.3 = 1.6$. This clearly shows that the optimal preorder structures are different for the two losses.

In the case of partial orders and when considering the space $\mathcal{R}_{par}$, situations where both labels are relevant or irrelevant are conflated, hence, we have

Table 1 $p(y_i, y_j)$,
 $1 \leq i < j \leq 3$

(y_1, y_2)	$p(y_1, y_2)$	(y_1, y_3)	$p(y_1, y_3)$	(y_2, y_3)	$p(y_2, y_3)$
(0,0)	0.60	(0,0)	0.10	(0,0)	0.30
(0,1)	0.25	(0,1)	0.70	(0,1)	0.00
(1,0)	0.10	(1,0)	0.10	(1,0)	0.00
(1,1)	0.05	(1,1)	0.10	(1,1)	0.70

$$p(\lambda_j \perp_R \lambda_i) := p_{i=j}^{i=j} := p(y^i = 0, y^j = 0) + p(y^i = 1, y^j = 1). \quad (21)$$

and Eq. (13) becomes

$$\hat{R} \in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \prod_{\lambda_i \succ_{\bar{R}} \lambda_j} p(\lambda_i \succ_{\bar{R}} \lambda_j) \prod_{\lambda_j \succ_{\bar{R}} \lambda_i} p(\lambda_j \succ_{\bar{R}} \lambda_i) \prod_{\lambda_j \perp_{\bar{R}} \lambda_i} p(\lambda_j \perp_{\bar{R}} \lambda_i) \quad (22)$$

For this case as well, there are instances for which a BOO (14) of the Hamming accuracy (11) may differ from a BOO (13) of the subset 0/1 accuracy (12). This is shown by the next example

Example 4 Assume the conditional probabilities (16), (17), and (21) are given in Table 2.

The collection of pairwise preference relations $\{y_1 \prec y_2, y_1 \perp y_3, y_2 \prec y_3\}$ has the highest joint conditional probability (22), i.e., $0.70 \cdot 0.60 \cdot 0.70 = 0.2940$ and the highest expected Hamming accuracy (14), i.e., $0.70 + 0.60 + 0.70 = 2.000$. However, it does not form a partial order because the transitivity (3) is not satisfied. For example, $r_{1,2} + r_{2,3} - 1 = 1 + 1 - 1 = 1 > r_{1,3} = 0$.

Again, with some calculation, we can verify that the only BOO (13) of the subset 0/1 accuracy (12) under the assumption of label independence is $\{y_1 \prec y_2, y_1 \perp y_3, y_2 \perp y_3\}$ with its probability (22) equal to $0.70 \cdot 0.60 \cdot 0.30 = 0.1260$. $\{y_1 \prec y_2, y_1 \prec y_3, y_2 \prec y_3\}$ is the second most probable preorder with the probability (22) $0.70 \cdot 0.25 \cdot 0.70 = 0.1225$. However, the only BOO (14) of the Hamming accuracy (11) is $\{y_1 \prec y_2, y_1 \prec y_3, y_2 \prec y_3\}$ with the expected Hamming accuracy (14) equals to $0.70 + 0.25 + 0.70 = 1.65$. $\{y_1 \prec y_2, y_1 \perp y_3, y_2 \perp y_3\}$ is the second most probable partial order in terms of expected Hamming accuracy (14), which equals $0.70 + 0.60 + 0.30 = 1.6$.

In practice, the pairwise probabilities can simply be estimated from the MLC observations. Detailed construction of the corresponding probabilistic classifiers is given in Appendix D.

4 Algorithmic Solutions

This section provides algorithmic solutions for the problem of finding BOOs (7), which is detailed in Sect. 3.3.2, under different metrics discussed in Sect. 3.2 based on integer linear programming. We would like to remind that whenever $R \in \mathcal{R} \in \{\mathcal{R}_{\text{pre}}, \mathcal{R}_{\text{par}}\}$, it should satisfy the transitivity property (3). As will be elaborated in this section, this can be ensured by customizing the constraints of integer linear programming.

In the following, we shall first show that, in the case $\mathcal{R} = \mathcal{R}_{\text{pre}}$, finding a BOO (20) of the subset 0/1 accuracy (12) under the independence of the pairwise preference relations (15) and finding a BOO (14) of the Hamming accuracy (11) can be done by

Table 2 $p(y_i, y_j)$,
 $1 \leq i < j \leq 3$

(y_1, y_2)	$p(y_1, y_2)$	(y_1, y_3)	$p(y_1, y_3)$	(y_2, y_3)	$p(y_2, y_3)$
(0,0)	0.10	(0,0)	0.60	(0,0)	0.30
(0,1)	0.70	(0,1)	0.25	(0,1)	0.70
(1,0)	0.10	(1,0)	0.15	(1,0)	0.00
(1,1)	0.00	(1,1)	0.00	(1,1)	0.00

solving integer linear programming problems. We then present similar results for the case $\mathcal{R} = \mathcal{R}_{\text{pre}}$. For both cases, the results can be easily extended to find BOOs of cost-sensitive Hamming accuracies as described in Appendix E.

Preorders

The next proposition defines an ILP whose result is the BOO for preorder structure and subset 0/1 accuracy (12).

Proposition 3 Let $\mathcal{R} = \mathcal{R}_{\text{pre}}$. Assume the independence of the pairwise preference relations (15). For each query instance $\mathbf{x}$, a BOO (20) of the subset 0/1 accuracy (12) is determined by $R := \{z_{i,j}^l \mid 1 \leq i < j \leq K, 1 \leq l \leq 4\}$, which is the solution of the following optimization problem:

$$\text{maximize } \sum_{1 \leq i < j \leq K} z_{i,j}^1 \log(\mathbf{p}_{i,j}^{1,0}) + z_{i,j}^2 \log(\mathbf{p}_{i,j}^{0,1}) + z_{i,j}^3 \log(\mathbf{p}_{i,j}^{0,0}) + z_{i,j}^4 \log(\mathbf{p}_{i,j}^{1,1}) \quad (23)$$

$$\text{subject to } \sum_{l=1}^4 z_{i,j}^l = 1, 1 \leq i < j \leq K, \quad (24)$$

$$z_{i,j}^l \in \{0, 1\}, 1 \leq l \leq 4, 1 \leq i < j \leq K, \quad (25)$$

$$r_{i,k} + r_{k,j} - 1 \leq r_{i,j}, \forall i \neq j \neq k. \quad (26)$$

The integer linear programming (23) can be easily adapted to find a BOO (14) of the Hamming accuracy (11). To find a BOO (14) of the Hamming accuracy (11), one needs to replace the target function of (23) with the following target function

$$\sum_{1 \leq i < j \leq K} (z_{i,j}^1 \mathbf{p}_{i,j}^{1,0}(\mathbf{x}) + z_{i,j}^2 \mathbf{p}_{i,j}^{0,1} + z_{i,j}^3 \mathbf{p}_{i,j}^{0,0} + z_{i,j}^4 \mathbf{p}_{i,j}^{1,1}). \quad (27)$$

As suggested in (Masson et al., 2016), it might also be interesting to restrict the search space to order structures of height 2. This shall be done by enlarging the set of constraints (24–26) with

$$r_{i,k} + r_{k,j} \leq 1, \forall i \neq j \neq k. \quad (28)$$

Since the constraints (28) imply the constraints (26), one can remove the (26).

Partial Orders

The next proposition defines an ILP whose result is the BOO for partial order structure and subset 0/1 accuracy (12).

Proposition 4 Let $\mathcal{R} = \mathcal{R}_{\text{par}}$. Assume the independence of the pairwise preference relations (22). For each query instance $\mathbf{x}$, a BOO (13) of the subset 0/1 accuracy (12) is determined by $R := \{z_{i,j}^l \mid 1 \leq i < j \leq K, 1 \leq l \leq 3\}$, which is the solution of the following optimization problem:

$$\text{maximize } \sum_{1 \leq i < j \leq K} z_{i,j}^1 \log(\mathbf{p}_{i,j}^{1,0}) + z_{i,j}^2 \log(\mathbf{p}_{i,j}^{0,1}) + z_{i,j}^3 \log(\mathbf{p}_{i,j}^{i=j}) \quad (29)$$

$$\text{subject to } \sum_{l=1}^3 z_{i,j}^l = 1, 1 \leq i < j \leq K, \quad (30)$$

$$z_{i,j}^l \in \{0, 1\}, 1 \leq l \leq 3, 1 \leq i < j \leq K, \quad (31)$$

$$r_{i,k} + r_{k,j} - 1 \leq r_{i,j}, \forall i \neq j \neq k. \quad (32)$$

To find a BOO (14) of the Hamming accuracy (11), one needs to replace the target function of (29) with the following target function

$$\sum_{1 \leq i < j \leq K} (z_{i,j}^1 \mathbf{p}_{i,j}^{1,0} + z_{i,j}^2 \mathbf{p}_{i,j}^{0,1} + z_{i,j}^3 \mathbf{p}_{i,j}^{i=j}). \quad (33)$$

Again, if one is only interested in predicting partial orders of height 2, we can replace the constraint (32) by (28).

Those ILP problems are NP-hard, as the linear ordering problem, which is a subcase of our formulation, is known to be NP-hard (Mitchell & Borchers, 2000). However, we will see in the experiments that open and free solvers such as GLPK (CVXOPT) and MILP (SciPy) can scale up to 53 labels, meaning that commercial ones such as Gurobi could go beyond. We refer to Appendix F.1 for preliminary results on the runtime of our proposals. This is sufficient for most applications, unless one wants to consider extreme multi-label problems (Babbar & Schölkopf, 2019) that anyway require very specific solutions. So our approach can accommodate most practical problems, but one should keep in mind the additional complexity, and balance it with the added benefits in terms of accuracy, interpretability, insights, etc., of having a richer structure in the prediction.

5 Experiments on the MLC Problem

While many of the results presented in this paper can readily be applied to learning problems involving order structures such as label ranking (Cheng et al., 2012), our goal here is to investigate how well they apply to MLC problems, notably in the presence of noise, imbalance, or when one allows for partial abstention. This is what we explore in detail in this section.

5.1 Extracting MLC Predictions from Orders

Once optimal order structures have been predicted, it remains to extract from them MLC predictions, as this is desirable to compare their performances with classical MLC approaches. Similar to what has been done in (Nguyen et al., 2018), we propose to extract a hard prediction from each given predicted preference order. More precisely, given a preference order R , which can be either a preorder or a partial order, for each label λ_k , $k = 1, \dots, K$, we predict

$$y_k = \begin{cases} 0 & \text{if } \sum_{j=1}^K z_{k,j}^1 = 0 \text{ and } \sum_{j=1}^K z_{k,j}^2 > 0 \\ 1 & \text{if otherwise.} \end{cases} \quad (34)$$

That is, we predict as irrelevant any label that does not strictly dominate any other, and is strictly dominated by at least one, and as relevant the other cases. This is illustrated in the next example.

Example 5 Figure 4 illustrates 2 generic order structures in the case $K = 7$ and 2 order structures of height 2 in the case $K = 6$.

The binary vector predictions extracted from the generic pre-order (c.f. Fig. 4a) and the generic partial order (c.f. Fig. 4b) are the same, i.e.,

$$(y^1, y^2, y^3, y^4, y^5, y^6, y^7) = (0, 0, 1, 1, 1, 1, 1). \quad (35)$$

The binary vector predictions extracted from the pre-order of height 2 (c.f. Fig. 4c) and the partial order of height 2 (c.f. Fig. 4d) are the same, i.e.,

$$(y^1, y^2, y^3, y^4, y^5, y^6) = (0, 1, 1, 1, 1, 1). \quad (36)$$

In our experiments, we will also consider MLC predictions with partial abstention (Nguyen & Hüllermeier, 2021). More precisely, given a preference order R , which can be either a preorder or a partial order, for each label λ_k , $k = 1, \dots, K$, we predict

$$y_k = \begin{cases} 0 & \text{if } \sum_{j=1}^K z_{k,j}^1 = 0 \text{ and } \sum_{j=1}^K z_{k,j}^2 > 0 \\ \mathbb{R} & \text{if } \sum_{j=1}^K z_{k,j}^1 = 0 \text{ and } \sum_{j=1}^K z_{k,j}^2 = 0 \\ 1 & \text{if otherwise.} \end{cases} \quad (37)$$

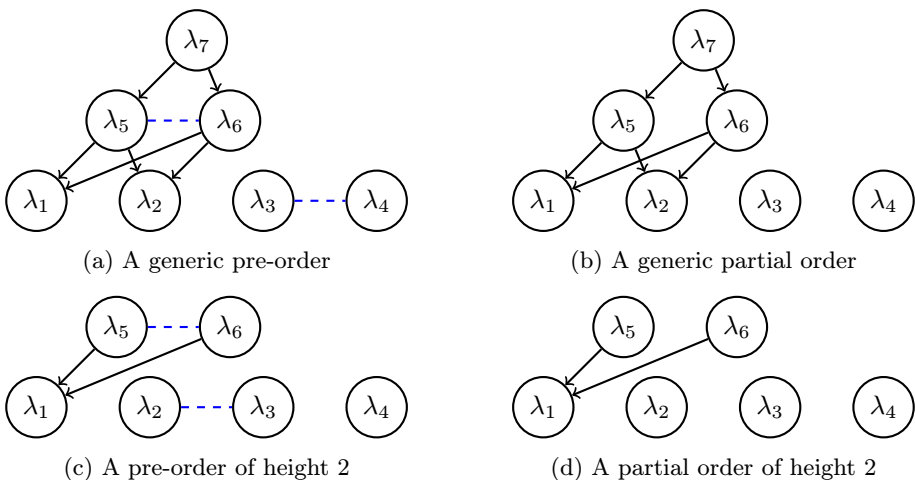


Fig. 4 Illustration of **a** a generic pre-order, **b** a generic partial order, **c** a pre-order of height 2, **d** a partial order of height 2. For pre-orders, $(\lambda_i \succ_R \lambda_j, \lambda_i \sim_R \lambda_j)$ and $\lambda_i \perp_R \lambda_j$ are symbolized by $(\lambda_i \rightarrow \lambda_j, \lambda_i - \lambda_j)$ and a missing edge, respectively. For partial orders, $\lambda_i \succ_R \lambda_j$ and $\lambda_i \perp_R \lambda_j$ are symbolized by $\lambda_i \rightarrow \lambda_j$ and a missing edge, respectively

In comparison to hard predictions, we abstain from labels that are incomparable with others. This way of extracting predictions with partial abstention is illustrated in the next example.

Example 6 We shall continue with the examples given in Fig. 4.

The predictions with partial abstention extracted from the generic pre-order (c.f. Fig. 4a) and the generic partial order (c.f. Fig. 4b) are the same, i.e.,

$$(y^1, y^2, y^3, y^4, y^5, y^6, y^7) = (0, 0, \mathbb{R}, \mathbb{R}, 1, 1, 1), \quad (38)$$

The predictions with partial abstention extracted from the pre-order of height 2 (c.f. Fig. 4c) and the partial order of height 2 (c.f. Fig. 4d) are the same, i.e.,

$$(y^1, y^2, y^3, y^4, y^5, y^6) = (0, \mathbb{R}, \mathbb{R}, \mathbb{R}, 1, 1), \quad (39)$$

Note that previous frameworks (Nguyen & Hüllermeier, 2021) with abstention were specially designed for scenarios where domain experts are available and provide domain-oriented rewards for being correctly cautious, while our proposal is especially designed as an alternative for scenarios where domain experts are not available.

5.2 Experimental Setting

We perform experiments on 10 data sets from <https://cometa.ujaen.es/datasets/>, which are summarized in Table 3. We follow a 5×5-fold cross-validation procedure and report the average scores of the metrics covered. MeanIR and CVIR (Tarekegn et al., 2021) are used to assess the imbalance level of the datasets. The imbalance ratio of each label is defined as the fraction of the relevance frequency of the most frequent label value over the frequency of the considered label: MeanIR is then the average imbalance ratio over the labels, while CVIR is the coefficient of variation of the imbalance ratios between labels. Higher ratios indicate more imbalanced situations.

Besides experiments on the original (clean) data sets, we shall also test with noisy data sets, where for each data set and each training data set, we shall swap the value of each label by tossing a biased coin with probability α . To generate different noisy situations, we shall conduct experiments with $\alpha \in \{0.1, 0.2, 0.3\}$. In the first set of experiments, varying the noise allows us to test the robustness of our approach compared with other classifiers. In the second set of experiments, varying the noise level allows us to test the precision-recall trade-off provided by the proposed classifiers.

As the quality of the probabilistic estimates $p_{i,j}$ is important to ensure good performance of our predictions and those of compared methods, we employ random forest (RF) (Ho, 1995), known to be competitive on tabular data (Grinsztajn et al., 2022), to produce pairwise probabilistic predictions. We use Sklearn's implementation with the default setting in all the experiments. We also conduct another set of experiments with LightGBM (Ke et al., 2017) as the base learner. We use the default setting <https://lightgbm.readthedocs.io/en/stable/>. The source code used in our experiments has been made public at <https://pypi.org/project/preorder4mlc/>.

Table 3 Data sets used in the experiments

#	Name	# of instances	# of features	# of labels	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	VirusPseAAC	207	440	6	8.62	0.92
3	CHD_49	555	49	6	5.77	1.60
4	Emotions	593	72	6	1.48	0.16
5	Scene	2407	294	6	1.25	0.11
6	PlantPseAAC	978	440	12	6.69	0.68
7	Water-quality	1060	16	14	1.77	0.29
8	Yeast	2417	103	14	7.20	1.82
9	HumanPseAAC	3106	440	14	15.29	1.05
10	Enron	1702	1001	53	10.45	0.80

5.2.1 Setting for Standard MLC Predictions

We first conduct a set of experiments to assess the potential applications of our proposal for predicting order structures in predicting binary vectors, i.e., the conventional MLC setting. This is motivated by the promising performance of CLR (Fürnkranz et al., 2008) that uses complete orders as structures. This set of experiments shall cover 8 proposed classifiers, which first predict order structures and then extract binary vectors, namely PR-H, PR-H-2, PR-S, PR-S-2, PA-H, PA-H-2, PA-S, and PA-S-2, and 3 MLC classifiers: CLR (Fürnkranz et al., 2008), binary relevance (BR) (Zhang et al., 2018), classifier chain (CC) (Read et al., 2021), and ensemble of CCs (ECC) with cardinality 10 (Read et al., 2021).

The acronym of each classifier encodes the framework we are considering:

- PA/PR: whether we use preorder or partial order as a predictive structure;
- H/S: whether we are optimizing the Hamming or the Subset 0/1 metric;
- 2: whether we limit the height of the order to 2 or not, i.e., whether we consider bipartite (partial) graphs or not.

The final MLC prediction is here obtained by using Eq. (34). As we want to study the potential benefits of our approach on imbalanced data, in addition to the $f_{\text{MLC}}^{\text{ham}}$ (9) and $f_{\text{MLC}}^{\text{sub}}$ (10), we also use the F_{β} measure (40) and Jaccard index (41)

$$f_{\text{MLC}}^{\beta}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{(1 + \beta^2) \sum_{k=1}^K y_k \hat{y}_k}{(1 + \beta^2) \sum_{k=1}^K y_k + \sum_{k=1}^K \hat{y}_k}, \quad (40)$$

$$f_{\text{MLC}}^{\text{Jac}}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\sum_{k=1}^K y_k \hat{y}_k}{\sum_{k=1}^K y_k + \sum_{k=1}^K \hat{y}_k - \sum_{k=1}^K y_k \hat{y}_k}. \quad (41)$$

which are commonly used to assess MLC predictions for imbalanced data sets. We also consider two balancing scores:

$$\text{MFRD}(\mathcal{T}) = \max_{k \in [K]} |\text{FPR}(Y^k | \mathcal{T}) - \text{FNR}(Y^k | \mathcal{T})| \quad (42)$$

$$\text{AFRD}(\mathcal{T}) = \frac{1}{K} \sum_{k=1}^K |\text{FPR}(Y^k | \mathcal{T}) - \text{FNR}(Y^k | \mathcal{T})|, \quad (43)$$

where $\mathcal{T}$ is the test data set, and for each $k \in [K]$,

$$\text{FPR}(Y^k | \mathcal{T}) = \frac{\text{FP}(Y^k | \mathcal{T})}{\text{FP}(Y^k | \mathcal{T}) + \text{TN}(Y^k | \mathcal{T})} \quad (44)$$

$$\text{FNR}(Y^k | \mathcal{T}) = \frac{\text{FN}(Y^k | \mathcal{T})}{\text{FN}(Y^k | \mathcal{T}) + \text{TP}(Y^k | \mathcal{T})} \quad (45)$$

and $\text{TP}(Y^k | \mathcal{T})$, $\text{FP}(Y^k | \mathcal{T})$, $\text{TN}(Y^k | \mathcal{T})$, and $\text{FN}(Y^k | \mathcal{T})$ are respectively the true positive, false negative, true negative and false negative rates on the predictions on Y^k on the test data set $\mathcal{T}$. Eqs. (42, 43) are low when the predictive powers are equally good for relevant and irrelevant labels. Moreover, we also assess the predictors using macro F_1 (46), and micro F_1 (47)

$$f_{\text{MLC}}^{\text{mac}}(\mathcal{T}) = \frac{1}{K} \sum_{k=1}^K \frac{2\text{TP}(Y^k | \mathcal{T})}{2\text{TP}(Y^k | \mathcal{T}) + \text{FN}(Y^k | \mathcal{T}) + \text{FP}(Y^k | \mathcal{T})}, \quad (46)$$

$$f_{\text{MLC}}^{\text{mic}}(\mathcal{T}) = \frac{\sum_{k=1}^K 2\text{TP}(Y^k | \mathcal{T})}{\sum_{k=1}^K (2\text{TP}(Y^k | \mathcal{T}) + \text{FN}(Y^k | \mathcal{T}) + \text{FP}(Y^k | \mathcal{T}))}, \quad (47)$$

where $\mathcal{T}$ is the test set.

5.2.2 Setting with Partial Abstention

In this set of experiments, we employ PR-H, PR-H-2, PR-S, PR-S-2, PA-H, PA-H-2, PA-S, and PA-S-2 to make predictions with partial abstention using Eq. (37). To check if allowing partial abstention may help improve the predictive performance, we assess these classifiers using the f_{MLC}^1 (40), $f_{\text{MLC}}^{\text{ham}}$ (9) and $f_{\text{MLC}}^{\text{sub}}$ (10) on the predicted labels. These metrics are complemented by the two metrics, which assess the abstention rates:

$$\text{AABS}(\mathcal{T}) = \frac{1}{K|\mathcal{T}|} \sum_{(\mathbf{x}_t, \mathbf{y}_t) \in \mathcal{T}} \sum_{k=1}^K \mathbb{I}[\hat{y}_t^k = \mathbb{R}], \quad (48)$$

$$\text{ABS}(\mathcal{T}) = \frac{1}{|\mathcal{T}|} \sum_{(\mathbf{x}_t, \mathbf{y}_t) \in \mathcal{T}} \mathbb{I}[\sum_{k=1}^K \mathbb{I}[\hat{y}_t^k = \mathbb{R}] \geq 1]. \quad (49)$$

AABS measures the average number of abstentions, while ABS measures the number of predictions for which we abstain (on at least one label).

5.3 Results

5.3.1 Standard MLC Predictions

The results given in Tables 4, 9, and 12, 13, 14, 15, 16, 17, 18, 19 in Appendix F suggest a few interesting trends. On clean data sets ($\alpha = 0.0$) and data sets with small to moderate noisy rates ($\alpha = 0.1$ and $\alpha = 0.2$), classifiers opting for partial orders, i.e., (PA-H-2, PA-H, PA-S-2, PA-S), and classifiers opting for preorders, i.e., (PR-H, PR-S), consistently outperform (BR, CC, CLR) in terms of f_{MLC}^1 (40), $f_{\text{MLC}}^{\text{jac}}$ (41) and balancing scores: AFRD (43) and MFRD (42). However, on data sets with a high noisy rate ($\alpha = 0.3$), only (PR-H, PR-S) can be capable of consistently outperforming (BR, CC, CLR) with respect to these metrics, which are all essentially balancing scores. A similar trend, but a bit less clear, is also observed for $f_{\text{MLC}}^{\text{mac}}$ (46), which is another balancing metric. As expected, ECC appears to be a rather accurate and robust competitor. It is interesting to see that classifiers opting for preorders and partial orders, especially (PR-H, PR-S) can outperform ECC on the majority of the tests. These clearly suggest the potential advantages of predicting order structures, in particular preorders, in the presence of noisy and/or imbalanced labels. This suggests that the higher expressiveness of preorders also translates into a practical advantage, and that (PR-H, PR-S) clearly appear to be preferable in our experiments.

Regarding the $f_{\text{MLC}}^{\text{mic}}$ (47), which takes into account the imbalance without distinguishing the labels, classifiers opting for pre-orders and partial orders either outperform or are competitive with (BR, CC, CLR, ECC) on the majority of the tests.

Our approaches perform less well for Hamming loss $f_{\text{MLC}}^{\text{ham}}$ (9) and subset loss $f_{\text{MLC}}^{\text{sub}}$ (10), yet the gain of BR, CC, CLR for $f_{\text{MLC}}^{\text{ham}}$ has to be put in perspective with their bad results on balancing scores, as high $f_{\text{MLC}}^{\text{ham}}$ is typically achieved by predicting a majority of 0, while subset scores are anyway very low.

Finally, in our experiments, CLR showed some interest in balanced data sets, e.g., Water quality (cf. Table 16), Emotions (cf. Table 12), and Scene (cf. Table 14), but might not be advantageous on (highly) imbalanced ones, except for CHD-49 (cf. Table 12). This may be due to either an unfavorable base learner for that classifier, which is RF in our experiments, or to more theoretical aspects (Mello et al., 2022) that remain to be investigated.

5.3.2 Predictions with Partial Abstentions

The results given in Tables 5, 10, 21, 22, 23, 24, 25, 26, 27, 28 in Appendix F show that, overall, making predictions with partial abstention usually lead to improvements on MLC scores, including f_{MLC}^1 (40), $f_{\text{MLC}}^{\text{ham}}$ (9), and $f_{\text{MLC}}^{\text{sub}}$ (10).

The abstention rates of classifiers using preorders are consistently lower than those of the classifiers predicting partial orders, however they can sometimes provide competitive improvements on MLC scores, in particular f_{MLC}^1 (40), $f_{\text{MLC}}^{\text{mac}}$ (46), $f_{\text{MLC}}^{\text{mic}}$ (47), and $f_{\text{MLC}}^{\text{ham}}$ (9). The average results regarding f_{MLC}^1 (40) in Table 29 in Appendix F.4 show that allowing predictions with partial abstention often leads to impressive improvement. Similar results

are observed for $f_{\text{MLC}}^{\text{jac}}$ (41) and are omitted. Again, this suggests that the higher expressiveness of preorders is advantageous. Also note that the classifiers that predict preorders have similar abstention rates, but PR-H and PR-S tend to be a bit more promising than PR-H-2

Table 4 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

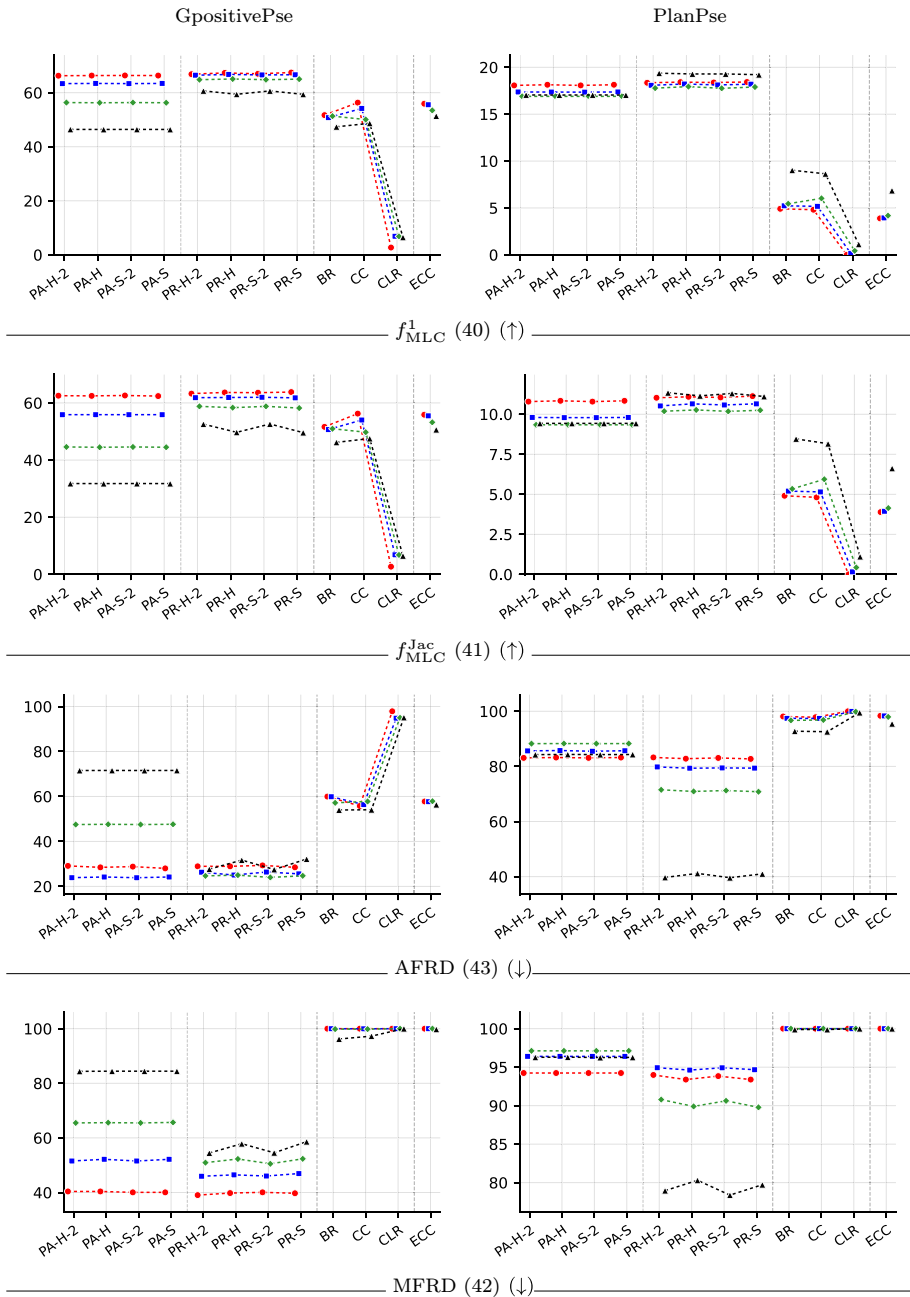
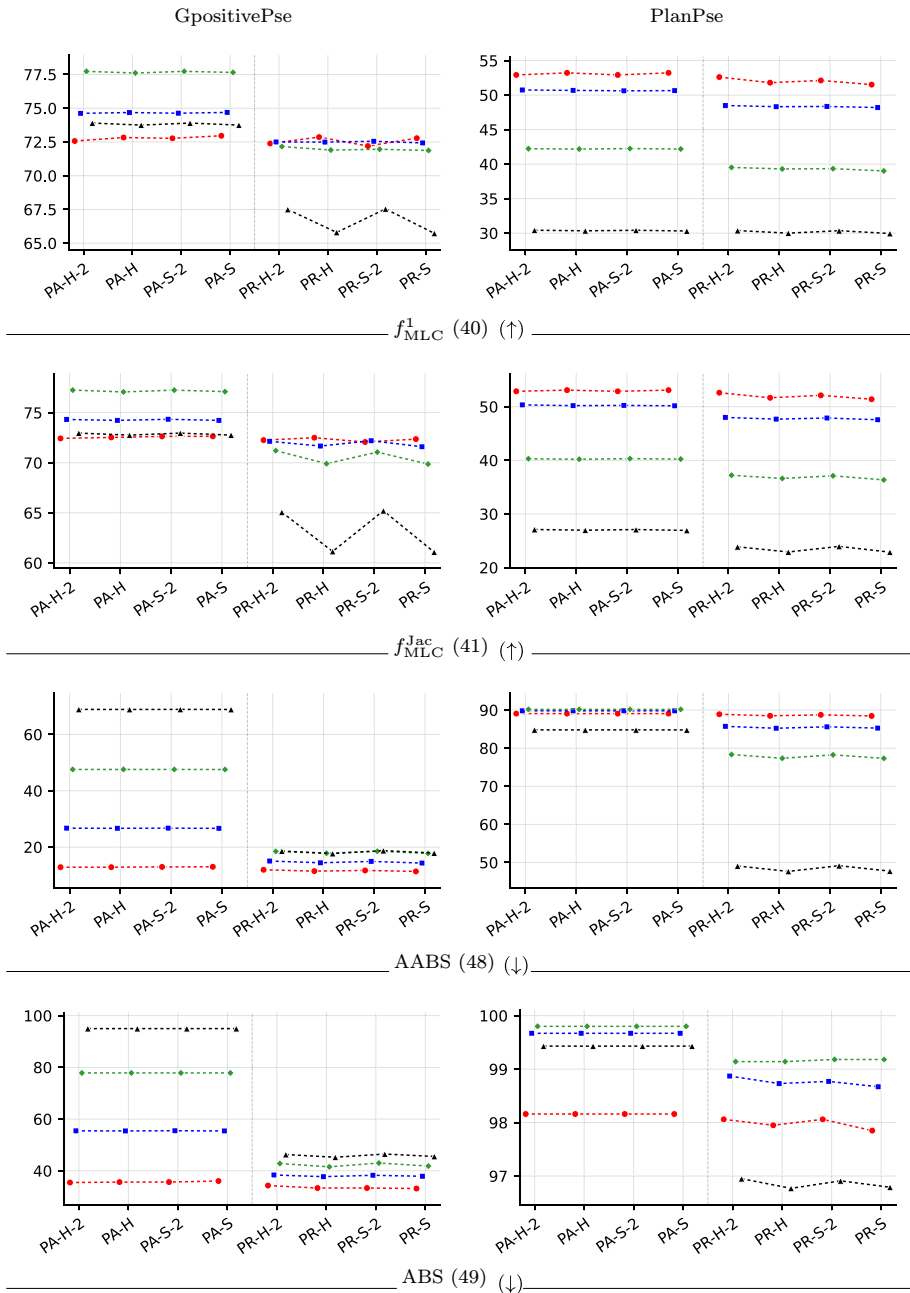


Table 5 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)



and PR-S-2 in improving the MLC scores. This is likely due to the fact that PR-H-2 and PR-S-2 limit the possibility of having incomparabilities and indifferences. Again, higher expressiveness also gives an empirical edge.

Finally, extracting predictions with partial abstention from order structures can lead to impressive improvements on MLC metrics on imbalanced data sets with higher numbers of labels, including plantPse and HumanPse (cf. Table 5). This is, however, sometimes combined with high abstention rates. So predicted order structures can effectively identify difficult labels (at the individual level) as isolated labels, which might in turn suggest meaningful notices to the end-user. However, it would be desirable to better control the abstention rate, e.g., by adapting existing multi-class techniques (Chzhen et al., 2021; Denis & Hebiri, 2017) to our own.

5.3.3 The Case of LightGBM as the Base Learner

Overall, classifiers opting for preorders, especially PR-H and PR-S, often outperform (BR, CLR, CC) and are on par with ECC. However, the gaps are less visible than in the case of random forests as the base learner. Moreover, allowing predictions with abstention can lead to promising improvements on MLC metrics, but again, less impressive than in the case of random forests as the base learner. We refer to Appendix F.4 for more discussions.

6 Conclusion

In this paper, we have proposed to cast the MLC problem into the problem of predicting order structures from probabilistic classifiers. We considered the corresponding issue of providing Bayes-optimal predictions in the form of constrained orders, exploring in particular preorders and partial orders with possible additional constraints. The advantage of using order structures is that they offer a highly expressive prediction space, allowing us to differentiate situations that would otherwise be considered equivalent. A prototypical example is the fact of considering two relevant labels as indifferent, and two irrelevant labels as incomparable.

Our empirical study demonstrates that our approaches are especially interesting in the case when one is interested in optimizing F-measures and when there is some imbalance in the observed labels, which commonly occurs within MLC problems.

Our proposal requires solving ILP problems to find the optimal solutions, and the current implementation using GLPK (CVXOPT) and MILP (SciPy), which are generic open and free solvers, can scale up to 53 labels. One might think of boosting the scalability and computational efficiency of the proposed framework by choosing stronger ILP solvers, such as the commercial Gurobi Optimizer.

While we focused here on the MLC setting with complete supervision, we would like to emphasize that our setting is quite general, in the sense that it can consider any problem where learning and predicting order structures make sense. In particular, we could explore our approach when considering label ranking problems (Cheng et al., 2012), or when considering multi-label problems with partial supervision (in which case incomparability can encode lack of information about label relevance). Finally, we would like to extend our approach by relaxing the independence assumption, for instance, by considering probabilities bearing on more than two labels.

Appendix A: Notation and Acronyms

Some frequently used notations and acronyms are listed in Table 6.

Table 6 Notation and acronyms

Symbol/Acronym	Meaning
$\mathcal{L} = \{\lambda_1, \dots, \lambda_K\}$	Set of labels
$\Lambda = \Lambda(\mathbf{x}) \in 2^{\mathcal{L}}$	Subset of relevant labels of $\mathbf{x}$
$\mathcal{X}, \mathbf{x}$	Instance space, instance
$\mathcal{Y} = \{0, 1\}^K, \mathbf{y}$	Set of possible labelings, labeling
$\mathcal{Y}_{\text{abs}} := \{0, 1, \textcircled{0}\}^K$	Set of possible predictions with partial abstention
$f_{\text{MLC}}, f_{\text{MLC}}^{\text{ham}}, f_{\text{MLC}}^{\text{sub}}, f_{\text{MLC}}^{\beta}, f_{\text{MLC}}^{\text{jac}}$	MLC metrics
$\mathbf{p}(\mathbf{y}) (\mathbf{p}(\mathbf{y} \mathbf{x})), \mathbf{p}(y^k) (\mathbf{p}(y^k \mathbf{x}))$	Conditional probabilities
$\lambda_i \succ_R \lambda_j (r_{i,j} = 1, r_{j,i} = 0)$	Strict preference
$\lambda_i \perp_R \lambda_j (r_{i,j} = 0, r_{j,i} = 0)$	Incomparability
$\lambda_i \sim_R \lambda_j (r_{i,j} = 1, r_{j,i} = 1)$	Indifference
$\llbracket \cdot \rrbracket$	The indicator function
$z_{i,j}^1 = \llbracket \lambda_i \succ_R \lambda_j \rrbracket = r_{i,j}(1 - r_{j,i})$	Value (0 or 1) of strict preference
$z_{i,j}^2 = \llbracket \lambda_j \succ_R \lambda_i \rrbracket = (1 - r_{i,j})r_{j,i}$	Value (0 or 1) of strict preference
$z_{i,j}^3 = \llbracket \lambda_i \perp_R \lambda_j \rrbracket = (1 - r_{i,j})(1 - r_{j,i})$	Value (0 or 1) of strict incomparability
$z_{i,j}^4 = \llbracket \lambda_i \sim_R \lambda_j \rrbracket = r_{i,j}r_{j,i}$	Value (0 or 1) of strict indifference
$\mathcal{R}$	Set of possible order structures
$\mathcal{R}_{\text{pre}}$	Set of possible preorders
$\mathcal{R}_{\text{par}}$	Set of possible partial orders
$f_{\text{OS}}, f_{\text{OS}}^{\text{ham}}, f_{\text{OS}}^{\text{sub}}, f_{\text{OS}}^{\beta}, f_{\text{OS}}^{\text{jac}}$	Metrics for order structures
$\mathbf{p}(\lambda_i \succ_R \lambda_j) := \mathbf{p}_{i,j}^{1,0} := \mathbf{p}(y_i = 1, y_j = 0)$	Conditional probabilities
$\mathbf{p}(\lambda_j \succ_R \lambda_i) := \mathbf{p}_{i,j}^{0,1} := \mathbf{p}(y_i = 0, y_j = 1)$	Conditional probabilities
$\mathbf{p}(\lambda_j \perp_R \lambda_i) := \mathbf{p}_{i,j}^{0,0} := \mathbf{p}(y_i = 0, y_j = 0)$	Conditional probabilities
$\mathbf{p}(\lambda_j \sim_R \lambda_i) := \mathbf{p}_{i,j}^{1,1} := \mathbf{p}(y_i = 1, y_j = 1)$	Conditional probabilities
$\mathbf{p}(R) (\mathbf{p}(R \mathbf{x})), \mathbf{p}(z_{i,j}^l) (\mathbf{p}(z_{i,j}^l \mathbf{x}))$	Conditional probabilities
MeanIR, CVIR	Imbalance ratios
MFRD, AFRD	Balancing scores
AABS, ABS	Abstention rates
MLC	Multi-label classification
BOP	Bayes-optimal prediction
BOO	Bayes-optimal order
BR	Binary relevance
CC	Classifier chains
CLR	Calibrated label ranking
RF	Random forest
PR-H, PR-H-2	Classifier outputs preorders optimizing $f_{\text{OS}}^{\text{ham}}$
PR-S, PR-S-2	Classifier outputs preorders optimizing $f_{\text{OS}}^{\text{sub}}$
PA-H, PA-H-2	Classifier outputs partial orders optimizing $f_{\text{OS}}^{\text{ham}}$
PA-S, PA-S-2	Classifier outputs partial orders optimizing $f_{\text{OS}}^{\text{sub}}$

Appendix B: Representation Capacity

In the following, we discuss the representation capacity of $\mathcal{R}_{\text{par}}$ and $\mathcal{R}_{\text{pre}}$ when being used to model the preference relations in the MLC setting. We shall also discuss the representation capacity of the CLR², which would be interpreted as either a linear order and a partial order of height 2 (or a bipartite graph) (Fürnkranz et al., 2008).

Partial Orders and Preorders

The framework for modeling and predicting partial orders from pairwise belief functions (Masson et al., 2016) is generic and can be easily adapted to the probabilistic setting by basically re-interpreting the pairwise scores as probabilistic scores. Once this is done, the resulting approach can be interpreted as an instantiation of our framework where $\mathcal{R} = \mathcal{R}_{\text{par}}$ and the BOO (7) is defined under the subset 0/1 accuracy (12) and the conditional probability is defined under the assumption of independence (Masson et al., 2016; Nguyen et al., 2018) on the pairwise preference relations. More precisely, for any $R \in \mathcal{R}_{\text{par}}$, we define $p(R | \mathbf{x})$ by using (22). The problem of finding a BOO (7) under different metrics is then followed by what has been investigated in Sect. 3.

We now illustrate some potential disadvantages of the use of partial order $\mathcal{R}_{\text{par}}$ as order structures in the MLC setting, where we only consider the relation $\lambda_i \perp_R \lambda_j$ rather than both $\lambda_i \sim_R \lambda_j$ and $\lambda_i \perp_R \lambda_j$. Intuitively, by considering that both the situations $(y_i, y_j) = (0, 0)$ and $(y_i, y_j) = (1, 1)$ correspond to the same relation, we tend to lower the chance of including strict preference (i.e., either $\lambda_i \succ_R \lambda_j$ or $\lambda_i \prec_R \lambda_j$). To illustrate this effect, let us take an example where $\mathcal{L} := \{\lambda_1, \lambda_2\}$ and

$$\begin{aligned} p_{1,2}^{1,0} &= 0.49, p_{1,2}^{0,1} = 0.00, \\ p_{1,2}^{0,0} &= 0.31, p_{1,2}^{1,1} = 0.20. \end{aligned}$$

In this case, the optimal preorder would be $\lambda_1 \succ \lambda_2$ with probability 0.49, while the optimal partial order is $\lambda_i \perp \lambda_j$ with the probability 0.51.

From an application perspective, it may also be quite useful to distinguish between the two situations $(y_i, y_j) = (0, 0)$ and $(y_i, y_j) = (1, 1)$. Let us consider the following (extreme) examples: if λ_1 and λ_2 encode two diseases, it might be reasonable to treat the two following patients (instances) differently

$$\begin{aligned} \text{Patient 1: } p_{1,2}^{1,0} &= p_{1,2}^{0,1} = p_{1,2}^{0,0} = 0, \quad p_{1,2}^{1,1} = 1, \\ \text{Patient 2: } p_{1,2}^{1,0} &= p_{1,2}^{0,1} = p_{1,2}^{1,1} = 0, \quad p_{1,2}^{0,0} = 1. \end{aligned}$$

If $\mathcal{R} = \mathcal{R}_{\text{pre}}$, one would clearly recommend the first patient to go to the hospital for further diagnoses, but not the second patient. However, if $\mathcal{R} = \mathcal{R}_{\text{par}}$, the two patients would receive the same recommendation/treatment.

²An interesting discussion about the robustness of CLR can be found in (Mello et al., 2022).

Calibrated Label Ranking

The output of CLR (Fürnkranz et al., 2008) can be interpreted as either linear orders or partial orders of height 2 (or bipartite graphs). Although the framework for learning BOOs proposed in Sect. 3 does not cover the approach for learning CLRs based on Ranking by pairwise comparison (RPC) procedure (Fürnkranz et al., 2008) as an instantiation, we would like to provide an extensive discussion on the notion of CLRs and the learning algorithm as it is one of a few existing preference learning approaches which are probabilistic and predict, for each instance, a partial order, i.e., closely related work to ours.

While the initial framework of (Fürnkranz et al., 2008) considered a ranking procedure, it also provides the option to turn the resulting complete ranking into a partial order of height 2 (or a bipartite graph) by estimating a threshold s_0 to separate the resulting ranking into two parts, called a set of relevant labels and a set of irrelevant labels, and ignore the strict preference relations $\lambda_i \succ_R \lambda_j$ for those pairs of labels fall into the same set (Fürnkranz et al., 2008).

By definition, the threshold s_0 is the expected number of irrelevant labels associated with $\mathbf{x}$, i.e.,

$$s_0 = \sum_{k=1}^K s_{0,k}^{1,0} = \sum_{k=1}^K \mathbf{p}(y^k = 0) = K - \sum_{k=1}^K \mathbf{p}(y^k = 1) = K - \mathbf{E}_{\mathbf{y}} \left(\sum_{k=1}^K y^k \right).$$

In practice, this step was done by employing a Binary Relevance (BR) classifier, i.e., to extract a collection of binary classification data sets, one per label, and train a set of binary classifiers $\{\mathbf{h}_{0,k} \mid k \in [K]\}$, one per binary classification data set. For each $\lambda_k \in \mathcal{L}$, the binary classifier $\mathbf{h}_{0,k}$ is trained on the training data set, which is created by annotating all the instances with $y^k = 0$ as positive and the remaining instances as negative.

Then, for each label λ_i , a ranking by pairwise comparison approach is employed to estimate its expected position in the ranking. More precisely, for each label pair (λ_i, λ_j) , CLR follows the learning by pairwise comparison procedure (Hüllermeier et al., 2008) and creates the data set $\mathcal{D}_{i,j}$ by annotating the instances within $\mathcal{D}$ using $c_{i,j}^{1,0}$, $c_{i,j}^{0,1}$, $c_{i,j}^{1,1}$ and $c_{i,j}^{0,0}$, which respectively encode the events $(y^i, y^j) = (1, 0)$, $(y^i, y^j) = (0, 1)$, $(y^i, y^j) = (0, 0)$, and $(y^i, y^j) = (1, 1)$. Then all the instances with the class $c_{i,j}^{1,1}$ and $c_{i,j}^{0,0}$ are eliminated from $\mathcal{D}_{i,j}$. Then any existing probabilistic classifier can be employed to learn $\mathbf{h}_{i,j}$, which produces pairwise information

$$(s_{i,j}^{1,0}, s_{i,j}^{0,1}) \text{ such that } s_{i,j}^{1,0} + s_{i,j}^{0,1} = 1. \quad (\text{B1})$$

This would be analogous to defining, for each label pair (λ_i, λ_j) , the pairwise information

$$s_{i,j}^{1,0} = \frac{p_{i,j}^{1,0}}{p_{i,j}^{1,0} + p_{i,j}^{0,1}}, s_{i,j}^{0,1} = \frac{p_{i,j}^{0,1}}{p_{i,j}^{1,0} + p_{i,j}^{0,1}}. \quad (\text{B2})$$

If we interpret the quantity (with abuse of notation)

$$\begin{aligned} s_i &= \sum_{j \neq i} s_{i,j}^{1,0} \approx \mathbb{E}_{\mathbf{y}} \left(\sum_{j \neq i} \llbracket \lambda_j \prec \lambda_i \rrbracket \right) = \sum_{\mathbf{y} \in \mathcal{Y}} \left(\sum_{j \neq i} \llbracket \lambda_j \prec \lambda_i \rrbracket \right) p(\mathbf{y}) \\ &= \sum_{j \neq i} \left(\sum_{\mathbf{y} \in \mathcal{Y}} \llbracket \lambda_j \prec \lambda_i \rrbracket p(\mathbf{y}) \right) = \sum_{j \neq i} p(y^i = 0, y^j = 1) = \sum_{j \neq i} p_{i,j}^{0,1}, \end{aligned}$$

i.e., the expected position of λ_i in the ranking (or the expected number of labels that are dominated by λ_i), CLR can be interpreted as a method that ranks the labels in decreasing order of the expected positions of the labels.

This would then suggest a nice way to separate relevant labels from irrelevant, as done in (Fürnkranz et al., 2008), i.e., a label $\lambda_i \in \mathcal{L}$ will be predicted as relevant if

$$s_i \approx \mathbb{E}_{\mathbf{y}} \left(\sum_{j \neq i} \llbracket \lambda_j \prec \lambda_i \rrbracket \right) \geq s_0 \approx \mathbb{E}_{\mathbf{y}} \left(\sum_{i=1}^K y^i \right). \quad (\text{B3})$$

Unfortunately, as shall be illustrated below, CLR may fail to capture the preference relations that differ from the strict preferences.

Let us take an example where

$$\begin{aligned} p_{1,2}^{1,0} &= 0.255, p_{1,2}^{0,1} = 0.255, \\ p_{1,2}^{0,0} &= 0.000, p_{1,2}^{1,1} = 0.490. \end{aligned}$$

Yet, the most probable preference relation should be $\lambda_1 \sim \lambda_2$. The CLR would return the prediction $(y^1, y^2) = (0, 0)$ (or the preference relation $\lambda_1 \perp \lambda_2$) because

$$\begin{aligned} s_1 &= s_{1,2}^{1,0} = \frac{p_{1,2}^{1,0}(\mathbf{x})}{p_{1,2}^{1,0} + p_{1,2}^{0,1}} = 0.5, \\ s_2 &= s_{1,2}^{0,1} = \frac{p_{1,2}^{0,1}(\mathbf{x})}{p_{1,2}^{1,0} + p_{1,2}^{0,1}} = 0.5, \end{aligned}$$

and $s_0 = (p_{1,2}^{1,0} + p_{1,2}^{0,0}) + (p_{1,2}^{0,1} + p_{1,2}^{0,0}) = 0.51$. As another example, let us take

$$\begin{aligned} p_{1,2}^{1,0} &= 0.01, p_{1,2}^{0,1} = 0.09, \\ p_{1,2}^{0,0} &= 0.05, p_{1,2}^{1,1} = 0.85. \end{aligned}$$

Clearly, the most probable preference relation should be $\lambda_1 \sim \lambda_2$. The CLR would return the prediction $(y^1, y^2) = (0, 1)$ (or the preference relation $\lambda_2 \succ \lambda_1$) because

$$s_1 = s_{1,2}^{1,0} = \frac{\mathbf{p}_{1,2}^{1,0}}{\mathbf{p}_{1,2}^{1,0} + \mathbf{p}_{1,2}^{0,1}} = 0.1,$$

$$s_2 = s_{1,2}^{0,1} = \frac{\mathbf{p}_{1,2}^{0,1}}{\mathbf{p}_{1,2}^{1,0} + \mathbf{p}_{1,2}^{0,1}} = 0.9,$$

and $s_0 = (\mathbf{p}_{1,2}^{1,0} + \mathbf{p}_{1,2}^{0,0}) + (\mathbf{p}_{1,2}^{0,1} + \mathbf{p}_{1,2}^{0,0}) = 0.2$.

Again, we see here some issues arising from the fact that CLR considers a framework with limited expressiveness, i.e., the space of complete rankings, when compared to ours.

Appendix C: Proofs of Formal Results

Proof of Proposition 1

A proof is simple and is given for completeness. By definition, we have

$$\hat{R} \in \hat{R}(\mathbf{x}) \in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} E_R(f_{\text{OS}}^{\text{sub}}(R, \bar{R})) = \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{R \in \mathcal{R}} \mathbb{I}[R = \bar{R}] \mathbf{p}(R) = \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \mathbf{p}(\bar{R}).$$

Proof of Proposition 2

A proof is again simple and is given for completeness. Let $T \in \{T_{\text{pre}}, T_{\text{par}}\}$ and $\mathcal{R} \in \{\mathcal{R}_{\text{pre}}, \mathcal{R}_{\text{par}}\}$. By definition, we have

$$\begin{aligned}
\hat{R} &\in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} E_R(f_{\text{OS}}^{\text{ham}}(R, \bar{R})) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{R \in \mathcal{R}} \left(\sum_{1 \leq i < j \leq K} \sum_{l=1}^L \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \right) \mathbf{p}(R) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \mathbf{p}(R) \right) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \mathbf{p}(T^{-1}(R)) \right) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \left(\sum_{\mathbf{y} \in T^{-1}(R)} \mathbf{p}(\mathbf{y}) \right) \right) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \left(\sum_{\mathbf{y} \in T^{-1}(R)} \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \mathbf{p}(\mathbf{y}) \right) \right) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \sum_{\mathbf{y} \in T^{-1}(R)} \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \mathbf{p}(\mathbf{y}) \right) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{\mathbf{y} \in \mathcal{Y}} \mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l \mathbf{p}(\mathbf{y}) \right) \\
&= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \bar{z}_{i,j}^l \mathbf{p}(\bar{z}_{i,j}^l),
\end{aligned}$$

The last equality holds because $\mathbb{I}[z_{i,j}^l = \bar{z}_{i,j}^l] z_{i,j}^l = 1$ iff $z_{i,j}^l = \bar{z}_{i,j}^l = 1$. Note that the map $T_{\text{pre}} : \mathcal{Y} \rightarrow T_{\text{pre}}(\mathcal{Y})$ is bijective and $\mathbf{p}(\bar{R}) = 0$ whenever $\bar{R} \notin T_{\text{pre}}(\mathcal{Y})$. Therefore, when $\mathcal{R} = \mathcal{R}_{\text{pre}}$, $|T_{\text{pre}}^{-1}(R)| = 1$, for any $R \in \mathcal{R}_{\text{pre}}$. However, when $\mathcal{R} = \mathcal{R}_{\text{par}}$, by definition, we have $|T_{\text{par}}^{-1}(R)| \geq 1$.

Proof of Proposition 3

We first reformulate the target function in (20) as

$$\mathbf{p}(R) = \prod_{1 \leq i < j \leq K} (\mathbf{p}_{i,j}^{1,0})^{z_{i,j}^1} (\mathbf{p}_{i,j}^{0,1})^{z_{i,j}^2} (\mathbf{p}_{i,j}^{0,0})^{z_{i,j}^3} (\mathbf{p}_{i,j}^{1,1})^{z_{i,j}^4}. \quad (\text{C4})$$

The constraint that, for any preorder $R \in \mathcal{R}$, each label pair (λ_i, λ_j) can only have one relation among $\lambda_i \succ_R \lambda_j$, $\lambda_j \succ_R \lambda_i$, $\lambda_i \sim_R \lambda_j$ and $\lambda_i \perp_R \lambda_j$ is translated into

$$\sum_{l=1}^4 z_{i,j}^l = 1, 1 \leq i < j \leq K. \quad (\text{C5})$$

Furthermore, the transitivity property

$$r_{i,k} + r_{k,j} - 1 \leq r_{i,j} \quad \forall i \neq j \neq k. \quad (\text{C6})$$

can be encoded using the relation $r_{i,j} = z_{i,j}^1 + z_{i,j}^4$ and $r_{j,i} = z_{j,i}^2 + z_{j,i}^4$ if $i < j$ and $j < i$, respectively. Once the constraints are ensured, we can solve the equivalent problem of optimizing the log version of (C4), which is more computationally convenient, compared to optimizing the original target function (C4).

Altogether, a most probable preorder (20) is determined by $\hat{R} := \{z_{i,j}^l \mid 1 \leq i < j \leq K, 1 \leq l \leq 4\}$, which is the solution of the optimization problem (23).

Proof of Proposition 4

We reformulate the target function in (22) as

$$p(R) = \prod_{1 \leq i < j \leq K} (p_{i,j}^{1,0})^{z_{i,j}^1} (p_{i,j}^{0,1})^{z_{i,j}^2} (p_{i,j}^{i=j})^{z_{i,j}^3}. \quad (\text{C7})$$

The constraint that, for any preorder $R \in \mathcal{R}$, each label pair (λ_i, λ_j) can only have one relation among $\lambda_i \succ_R \lambda_j$, $\lambda_j \succ_R \lambda_i$, $\lambda_i \sim_R \lambda_j$ and $\lambda_i \not\succ_R \not\prec \lambda_j$ is translated into

$$\sum_{l=1}^3 z_{i,j}^l = 1, 1 \leq i < j \leq K. \quad (\text{C8})$$

Furthermore, the transitivity property

$$r_{i,k} + r_{k,j} - 1 \leq r_{i,j}, \forall i \neq j \neq k. \quad (\text{C9})$$

can be encoded using the relation $r_{i,j} = 1 - z_{i,j}^2 - z_{i,j}^3$ and $r_{j,i} = 1 - z_{i,j}^1 - z_{i,j}^3$ if $i < j$ and $j < i$, respectively. Once the constraints are ensured, we can solve the equivalent problem of optimizing the log version of (C7).

Altogether, a most probable partial order (22) is determined by $\hat{R} := \{z_{i,j}^l \mid 1 \leq i < j \leq K, 1 \leq l \leq 3\}$, which is the solution of the optimization problem (29).

Appendix D: Estimating Conditional Probabilities

Preorders

When $\mathcal{R} \subseteq \mathcal{R}_{\text{pre}}$, we use, for each label pair (λ_i, λ_j) , a probabilistic 4-class classifier, to estimate the conditional probabilities (16–19) from the training data.

Inspired by what has been done in (Fürnkranz et al., 2008), for each label pair (λ_i, λ_j) , we use the learning by pairwise comparison procedure (Hüllermeier et al., 2008) and create the data set $\mathcal{D}_{i,j}$ by annotating the instances within $\mathcal{D}$ using $c_{i,j}^{1,0}$, $c_{i,j}^{0,1}$, $c_{i,j}^{1,1}$ and $c_{i,j}^{0,0}$. We then train a probabilistic 4-class classifier $h_{i,j} : \mathcal{X} \rightarrow \{c_{i,j}^{1,0}, c_{i,j}^{0,1}, c_{i,j}^{0,0}, c_{i,j}^{1,1}\}$, which predicts, for each query instance x :

$$(p_{i,j}^{1,0}, p_{i,j}^{0,1}, p_{i,j}^{0,0}, p_{i,j}^{1,1}) = (p(c_{i,j}^{1,0}), p(c_{i,j}^{0,1}), p(c_{i,j}^{0,0}), p(c_{i,j}^{1,1})) .$$

Partial Orders

When $\mathcal{R} \subseteq \mathcal{R}_{\text{par}}$, we use, for each label pair (λ_i, λ_j) , a probabilistic 3-class classifier, to estimate the conditional probabilities (16, 17, and 21) from the training data.

For each label pair (λ_i, λ_j) , we use the learning by pairwise comparison procedure (Hüllermeier et al., 2008) and create the data set $\mathcal{D}_{i,j}$ by annotating the instances within $\mathcal{D}$ using $c_{i,j}^{1,0}$, $c_{i,j}^{0,1}$, and $c_{i,j}^{i=j}$, where $c_{i,j}^{i=j}$ encode the event $y^i = y^j$. We then train a probabilistic 3-class classifier $h_{i,j} : \mathcal{X} \rightarrow \{c_{i,j}^{1,0}, c_{i,j}^{0,1}, c_{i,j}^{i=j}\}$, which predicts, for each query instance x :

$$(p_{i,j}^{1,0}, p_{i,j}^{0,1}, p_{i,j}^{i=j}) = (p(c_{i,j}^{1,0}), p(c_{i,j}^{0,1}), p(c_{i,j}^{i=j})) .$$

Appendix E: Extension to Other Metrics

Cost-Sensitive Hamming Accuracies

We can generalize the Hamming accuracy (11) as a family of cost-sensitive Hamming accuracies as follows:

$$f_{\text{OS}}^{\text{cs-ham}}(R, \hat{R}) = \sum_{1 \leq i < j \leq K} \sum_{l=1}^L f_{i,j}(z_{i,j}^l, \hat{z}_{i,j}^l) z_{i,j}^l, \quad (\text{E10})$$

where $L = 4$ and 3 for $\mathcal{R}_{\text{pre}}$ and $\mathcal{R}_{\text{par}}$, respectively, and $f_{i,j}(\cdot, \cdot)$ can take 4 possibly different values, one per pair $(z_{i,j}^l, \hat{z}_{i,j}^l) \in \{0, 1\}^2$. This can be seen as a direct extension of cost-sensitive accuracies for the binary classification task (Elkan, 2001).

We have the following rather direct result.

Proposition 5 Let $\mathcal{R} \in \{\mathcal{R}_{\text{pre}}, \mathcal{R}_{\text{par}}\}$. For each query instance x , a BOO (7) of the cost-sensitive Hamming accuracy (E10) is an order structure $\hat{R} \in \mathcal{R}$ with the highest expected cost-sensitive Hamming accuracy, i.e.,

$$\hat{R} \in \operatorname{argmax}_{\hat{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \sum_{v_{i,j}^l \in \{0,1\}} v_{i,j}^l f_{i,j}(v_{i,j}^l, \hat{z}_{i,j}^l) p(v_{i,j}^l), \quad (\text{E11})$$

with $L = 4$ and 3 for $\mathcal{R}_{\text{pre}}$ and $\mathcal{R}_{\text{par}}$, respectively.

Proof By definition, we have

$$\begin{aligned}
 \hat{R} &\in \operatorname{argmax}_{\bar{R} \in \mathcal{R}} E_R(f_{\text{OS}}^{\text{cs-ham}}(R, \bar{R})) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{R \in \mathcal{R}} \left(\sum_{1 \leq i < j \leq K} \sum_{l=1}^L f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l \right) p(R) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l p(R) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l p(T^{-1}(R)) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l \left(\sum_{\mathbf{y} \in T^{-1}(R)} p(\mathbf{y}) \right) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \left(\sum_{\mathbf{y} \in T^{-1}(R)} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l p(\mathbf{y}) \right) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{R \in \mathcal{R}} \sum_{\mathbf{y} \in T^{-1}(R)} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l p(\mathbf{y}) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{\mathbf{y} \in \mathcal{Y}} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l p(\mathbf{y}) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{v_{i,j}^l \in \{0,1\}} \left(\sum_{\substack{\mathbf{y} \in \mathcal{Y} \\ z_{i,j}^l = v_{i,j}^l}} f_{i,j}(z_{i,j}^l, \bar{z}_{i,j}^l) z_{i,j}^l p(\mathbf{y}) \right) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{v_{i,j}^l \in \{0,1\}} v_{i,j}^l f_{i,j}(v_{i,j}^l, \bar{z}_{i,j}^l) \left(\sum_{\substack{\mathbf{y} \in \mathcal{Y} \\ z_{i,j}^l = v_{i,j}^l}} p(\mathbf{y}) \right) \right) \\
 &= \operatorname{argmax}_{\bar{R} \in \mathcal{R}} \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \left(\sum_{v_{i,j}^l \in \{0,1\}} v_{i,j}^l f_{i,j}(v_{i,j}^l, \bar{z}_{i,j}^l) p(v_{i,j}^l) \right).
 \end{aligned}$$

□

Clearly, the target function of (23) should be replaced with

$$\sum_{1 \leq i < j \leq K} \left(\sum_{l=1}^L \left(\sum_{v_{i,j}^l \in \{0,1\}} f_{i,j}(v_{i,j}^l, z_{i,j}^l) v_{i,j}^l p(v_{i,j}^l) \right) \right) \quad (\text{E12})$$

when finding a BOO of any cost-sensitive Hamming accuracy of the form (E10).

Other Metrics

One might also think of generalizing the so-called family of confusion matrix-derived accuracies (Nguyen & Hüllermeier, 2021; Powers, 2011), which basically gives different rewards for being correct in predicting relevant labels and irrelevant labels, to our setting.

This would be done by first introducing the notion of relevant binary relations, denoted by $\mathcal{Z}_{\text{OS}} \subset \{z_{i,j}^l \mid 1 \leq l \leq L, 1 \leq i < j \leq K\}$. Once $\mathcal{Z}_{\text{OS}}$ is specified, for each pair of order structures $(R, \hat{R})$, one can construct the confusion matrix by replacing the notions of true positives (negatives) and false positives (negatives) by true relevances (irrelevances) and false relevances (irrelevances), respectively. This would allow one to generalize various confusion matrix-derived accuracies to our setting of predicting order structures.

As an illustration, the f_{MLC}^β -measure (40) would be generalized to the setting of preference learning as follows:

$$f_{\text{OS}}^\beta(R, \hat{R}) = \frac{(1 + \beta^2) \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \mathbb{I}[z_{i,j}^l \in \mathcal{Z}_{\text{OS}}] \mathbb{I}[z_{i,j}^l = \hat{z}_{i,j}^l]}{(1 + \beta^2) \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \mathbb{I}[z_{i,j}^l \in \mathcal{Z}_{\text{OS}}] + \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \mathbb{I}[\hat{z}_{i,j}^l \in \mathcal{Z}_{\text{OS}}]}. \quad (\text{E13})$$

Similarly, the Jaccard index (41) would also be generalized to the setting of preference learning as follows:

$$f_{\text{OS}}^{\text{Jac}}(R, \hat{R}) = \frac{g_{\text{OS}}^{\text{Jac}}(R, \hat{R})}{\sum_{1 \leq i < j \leq K} \sum_{l=1}^L (\mathbb{I}[z_{i,j}^l \in \mathcal{Z}_{\text{OS}}] + \mathbb{I}[\hat{z}_{i,j}^l \in \mathcal{Z}_{\text{OS}}]) - g_{\text{OS}}^{\text{Jac}}(R, \hat{R})}, \quad (\text{E14})$$

where

$$g_{\text{OS}}^{\text{Jac}}(R, \hat{R}) = \sum_{1 \leq i < j \leq K} \sum_{l=1}^L \mathbb{I}[z_{i,j}^l \in \mathcal{Z}_{\text{OS}}] \mathbb{I}[z_{i,j}^l = \hat{z}_{i,j}^l]. \quad (\text{E15})$$

However, the computational complexity of finding BOOs (7) of f_{OS}^β (E13), $f_{\text{OS}}^{\text{Jac}}$ (E14), and other confusion matrix-derived accuracies remains to be investigated.

Reminding that in the standard MLC setting, the divide-and-conquer strategy, which partitions the space of predictions into different groups whose local optimal solutions can be found efficiently by sorting the labels, has been employed to construct polynomial-time algorithms to find BOPs of the MLC F_β -measure (40) in the general setting of label dependence (Waegeman et al., 2014) and various confusion matrix-derived accuracies under the

independence assumption (Nguyen & Hüllermeier, 2021). However, as discussed in Sect. 3.3.2, even under the independence assumption, just optimizing (7) in a pairwise way will generally not produce an order structure. Therefore, readers who might be interested in generalizing such algorithms to our setting should ensure the transitivity (3) when determining the local optimal solutions.

Appendix F: Experimental Results

A Preliminary Study on the Runtime

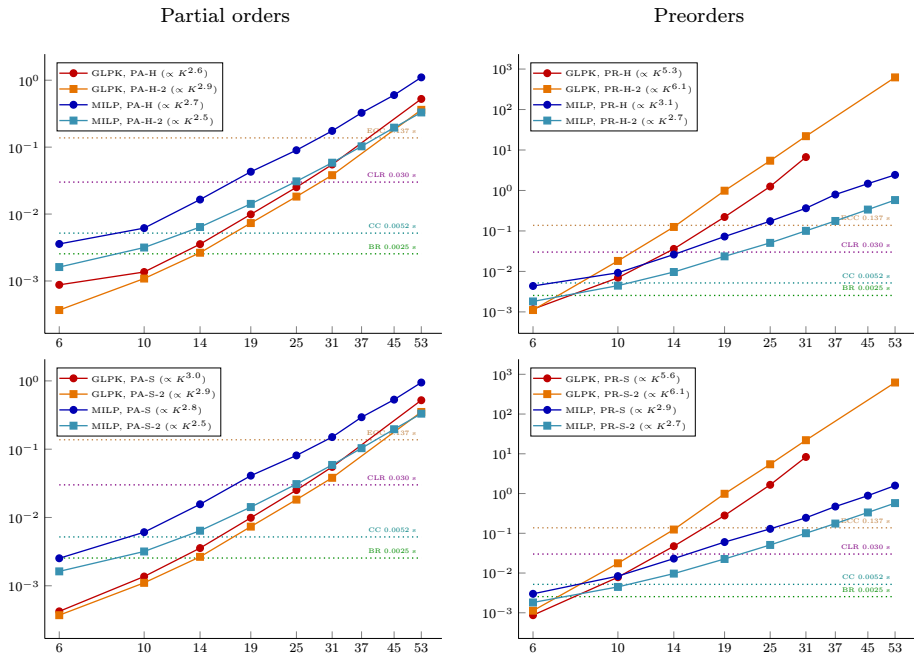
This section provides preliminary results on the runtime of our proposals when implemented using GLPK (CVXOPT) and MILP (SciPy), which are generic, open, and free ILP solvers.

The runtime of our proposals and competitors, i.e., (BR, CC, CLR, ECC), on the Enron data sets with 53 labels is given in Table 7. The results are computed using 300 randomly selected instances, except for when PR-H-2 and PR-S-2 are implemented using GLPK, where the results are computed using 10 randomly selected instances. n/a[†] means that the run-time exceeds the budget of our computing server. When implemented with MILP, classifiers that opt for pre-orders and partial orders require at most 2.430 seconds to search for optimal order structures.

To see how runtime varies as the number K of labels increases, we compute runtime for synthetic MLC problems constructed from the Enron data set. For each number $K \in \{6, 10, 14, 19, 25, 31, 37, 45\}$, we randomly select a subset $\mathcal{L}_K \subset \mathcal{L}$, which consists of the 53 labels, to form a synthetic MLC problem with K labels. Then we randomly select a subset of instances and compute the average per-instance prediction time as done for the case $K = 53$, which is reported in Table 7.

Table 7 Average per-instance prediction time (in seconds) on Enron

Classifier	GLPK	MILP
PA-H-2	0.361	0.330
PA-H	0.525	1.100
PA-S-2	0.351	0.331
PA-S	0.521	0.947
PR-H-2	622.64	0.579
PR-H	n/a [†]	2.430
PR-S-2	621.09	0.575
PR-S	n/a [†]	1.590
Competitors		
BR	0.0025	
CC	0.0052	
CLR	0.0300	
ECC	0.1370	

Table 8 Average per-instance prediction time (in seconds): Synthetic data sets with $K \in \{6, 10, 14, 19, 25, 31, 37, 45, 53\}$ 

The results given in Table 8 suggest that the runtime of both GLPK and MILP exhibits polynomial-like trends. This would be partly due to the fact that our ILP problems have sparse constraint matrices. Moreover, GLPK and MILP benefit classifiers opting for partial orders, i.e., (PA-H-2, PA-H, PA-S-2, PA-S), and classifiers opting for pre-orders, i.e., (PR-H-2, PR-H, PR-S-2, PR-S), respectively.

Additional Results for Section 5

- **Predicting binary vectors:** Additional results for GpositivePse and PlanPse are given in Table 9.
- **Predictions with partial abstention:** Additional results for GpositivePse and PlanPse are given in Table 10.

Table 9 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

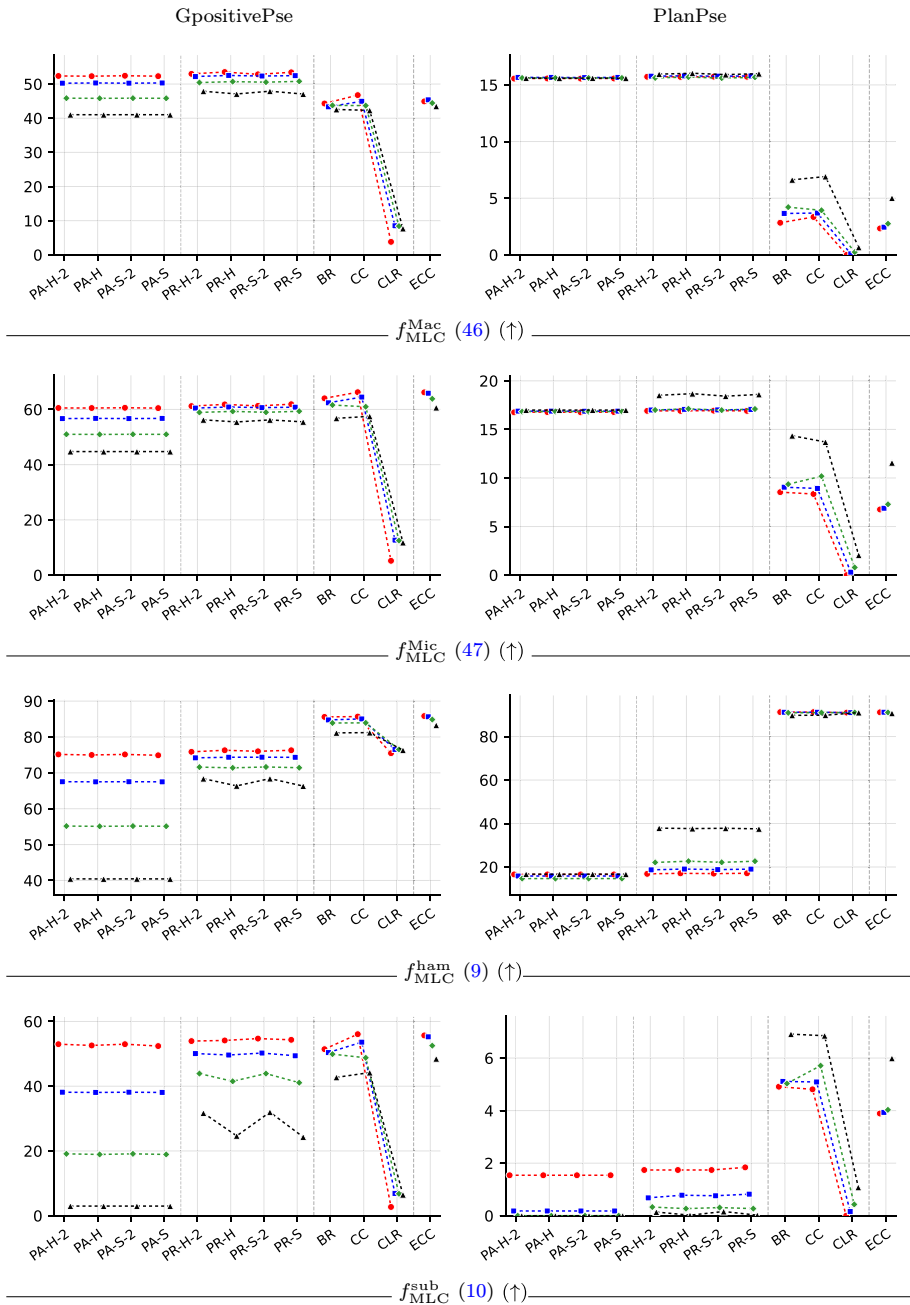
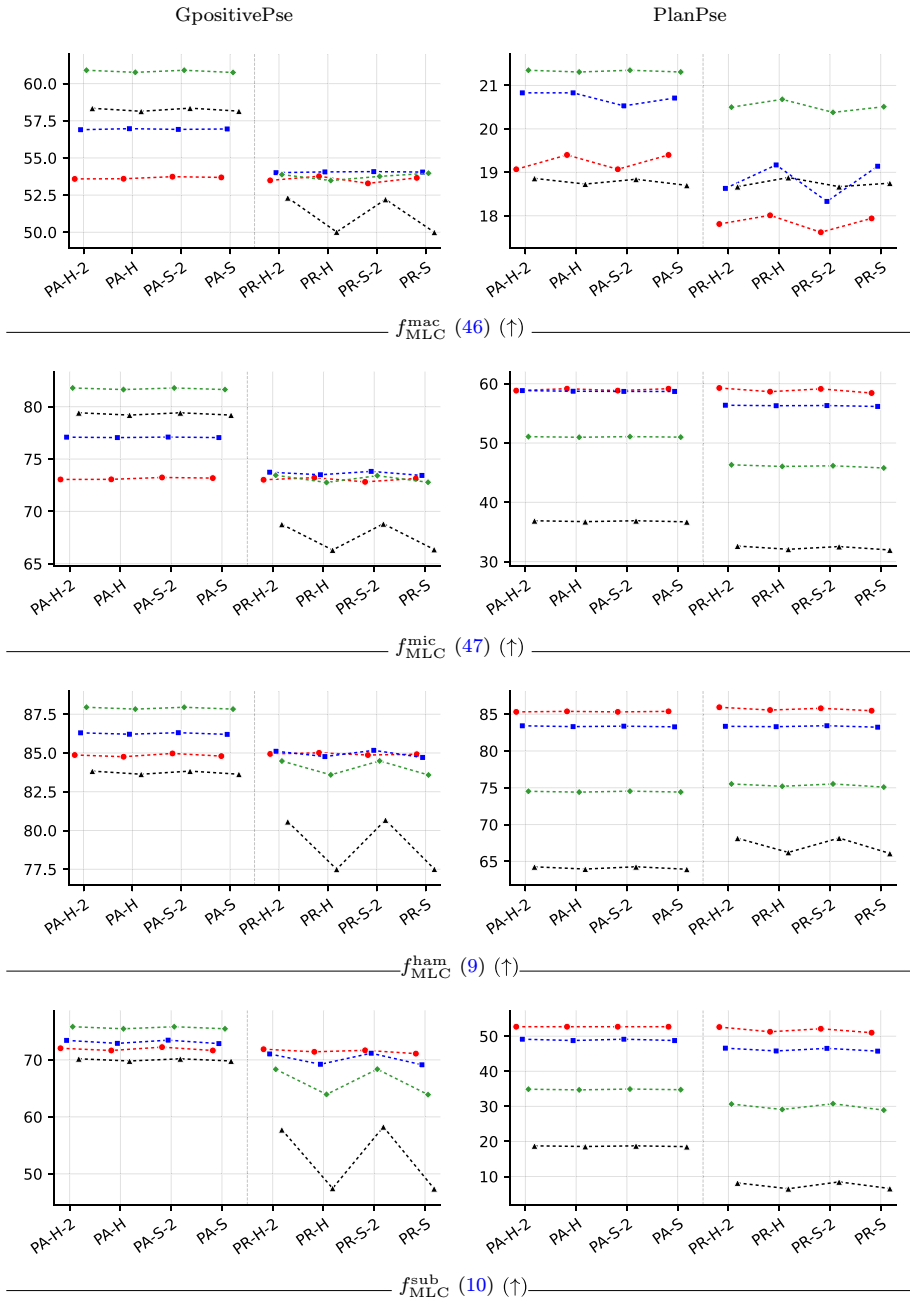


Table 10 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)



Results for Other Data Sets

Predicting Binary Vectors

A summary of the results for predicting binary vectors is given in Table 11.
See Tables 12, 13, 14, 15, 16, 17, 18 and 19.

Table 11 Predicting binary vectors: names of datasets and the tables with corresponding results

Names	Place
GpositivePseAAC & PlantPseAAC	Tables 4 and 9
CHD_49 & Emotions	Tables 12 and 13
VirusPseAAC & Scene	Tables 14 and 15
Water-quality & Yeast	Tables 16 and 17
HumanPseAAC & Enron	Tables 18 and 19

Table 12 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

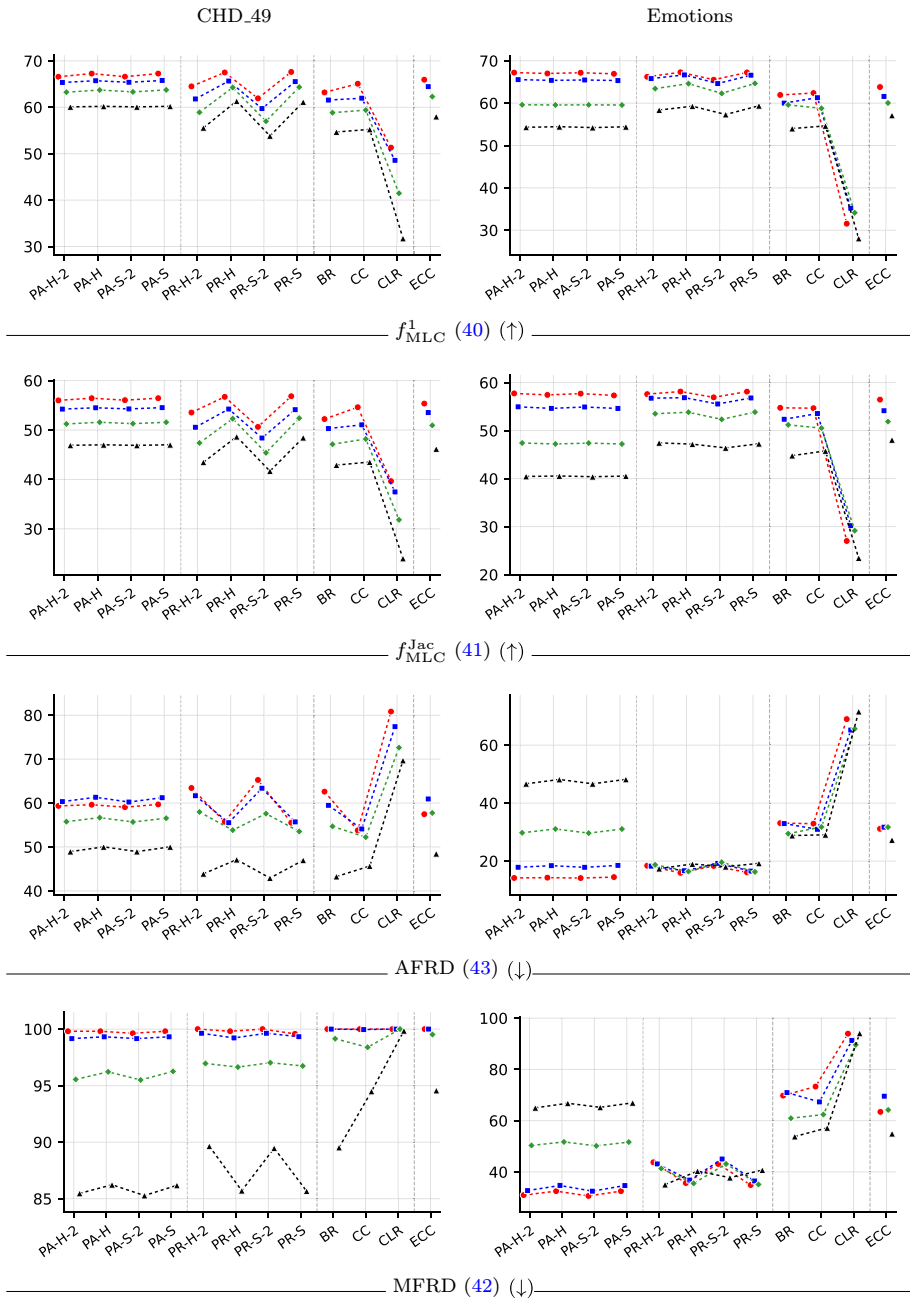


Table 13 Average scores (in%, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

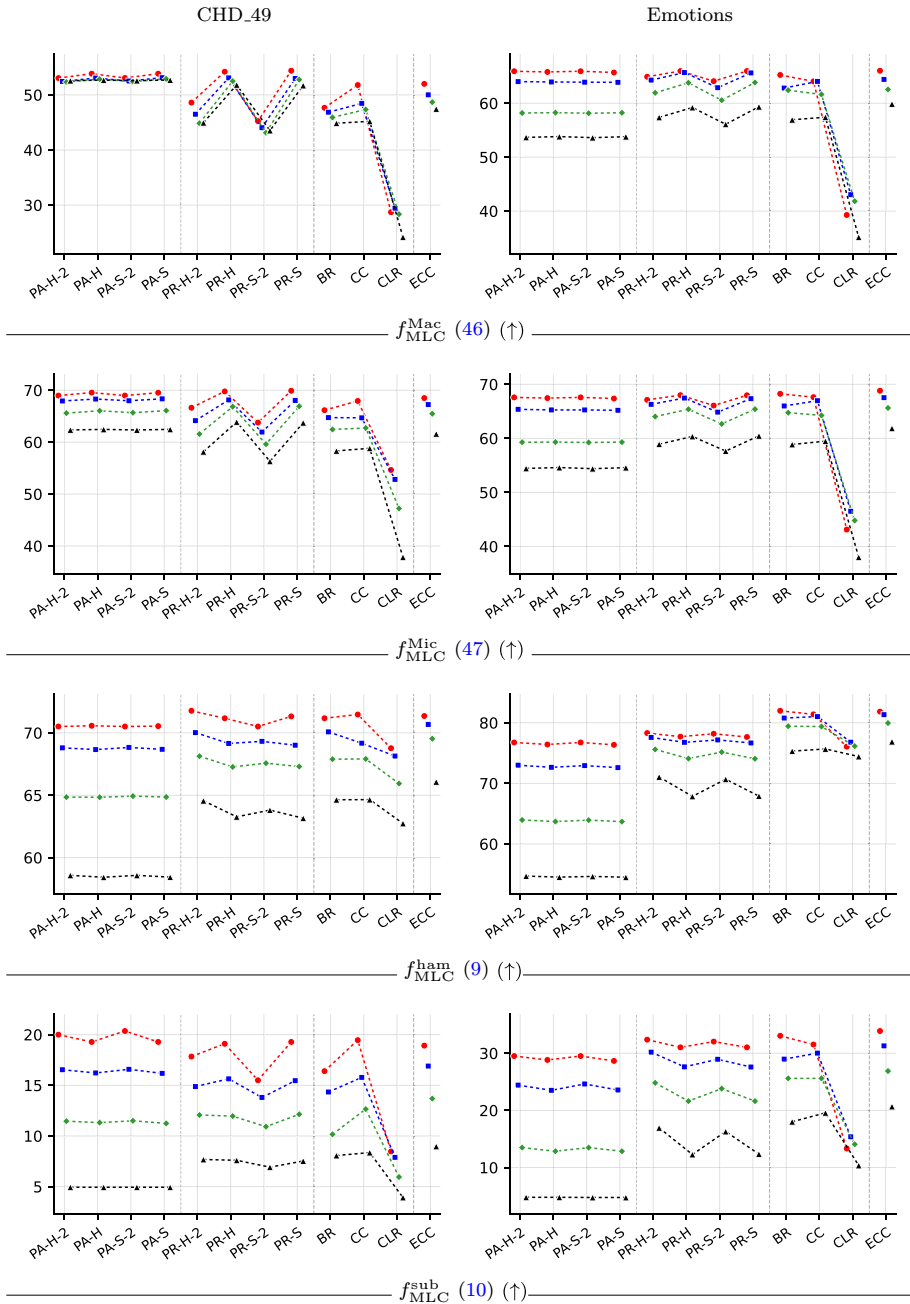


Table 14 Average scores (in%, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

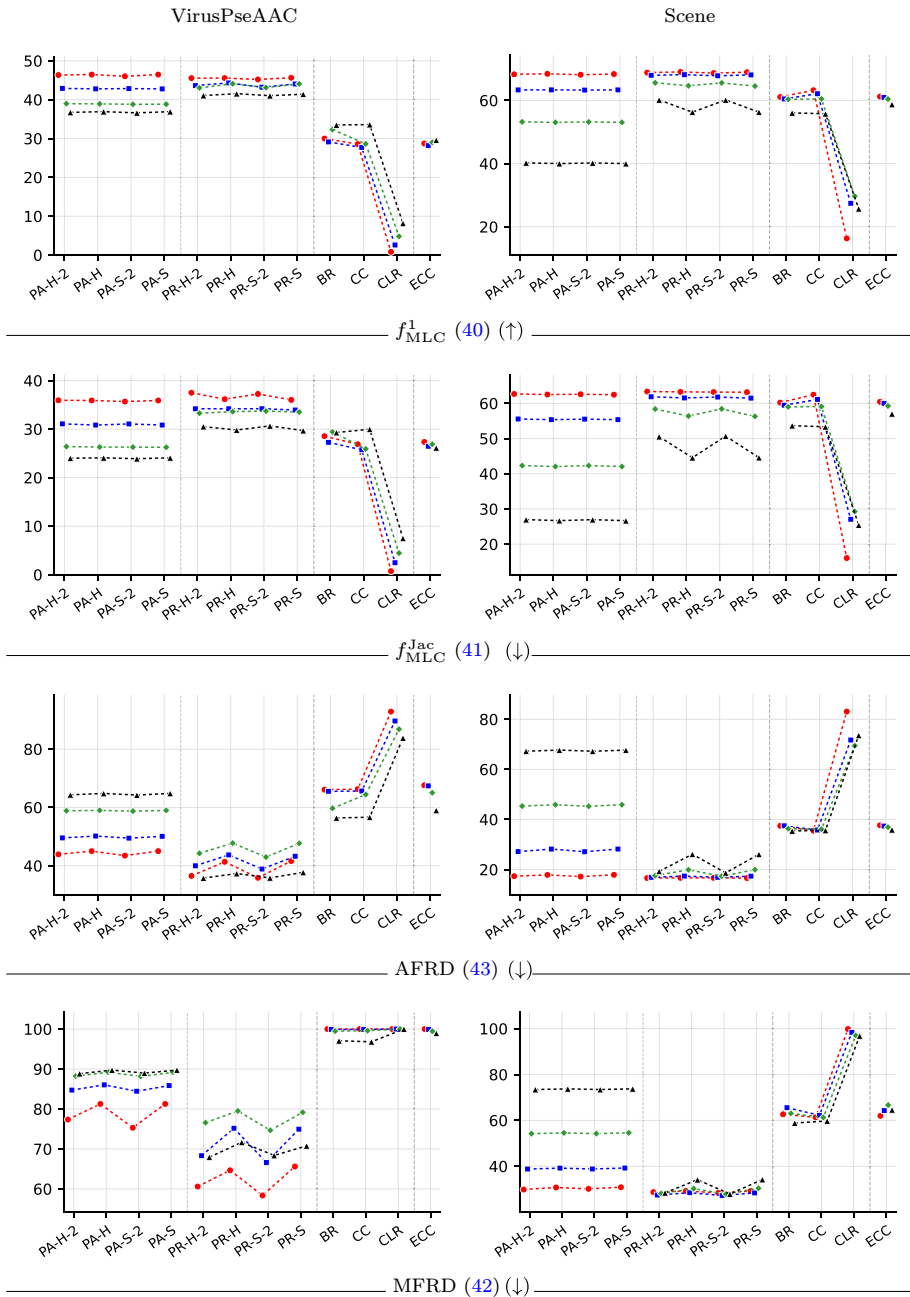


Table 15 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

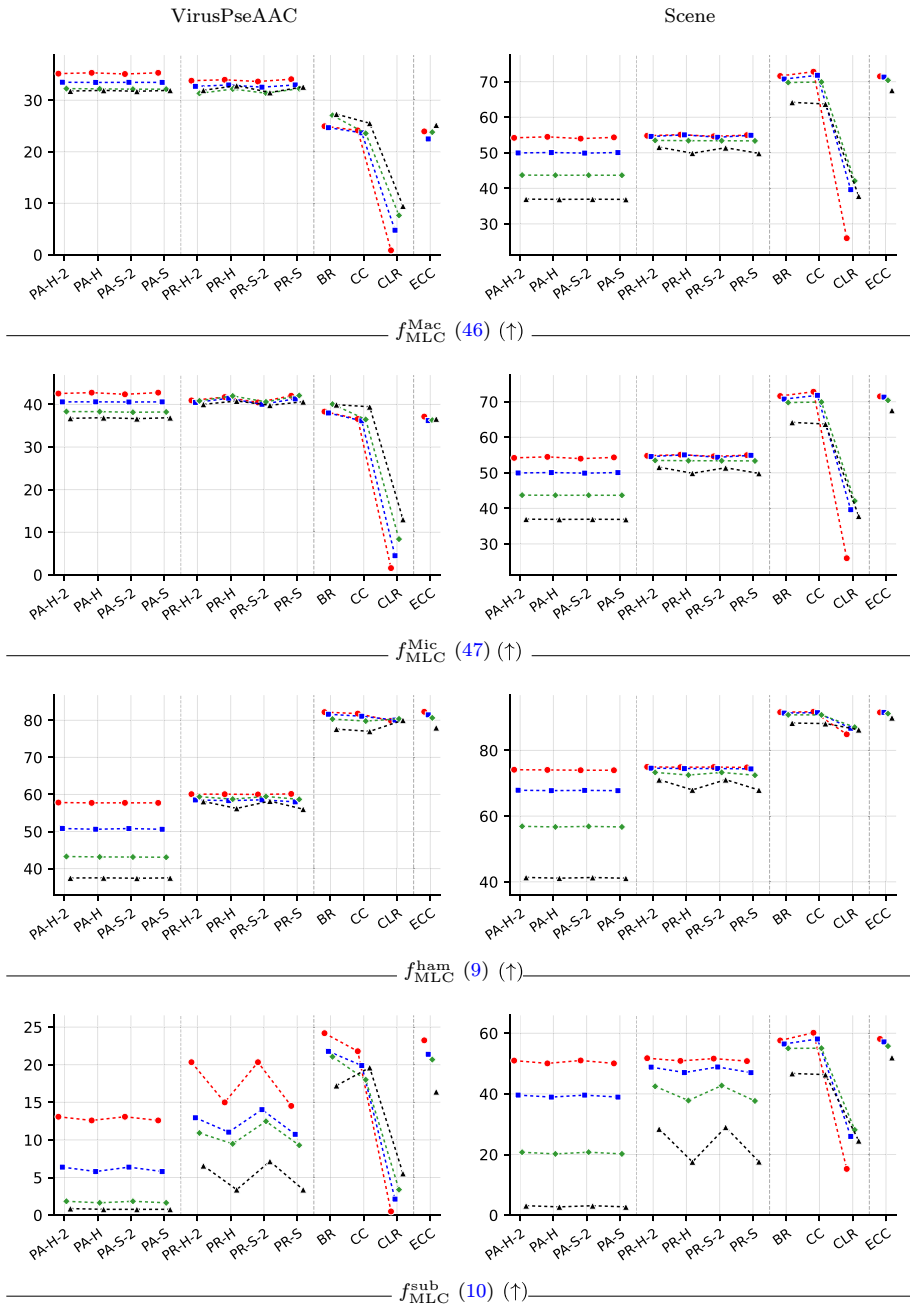


Table 16 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

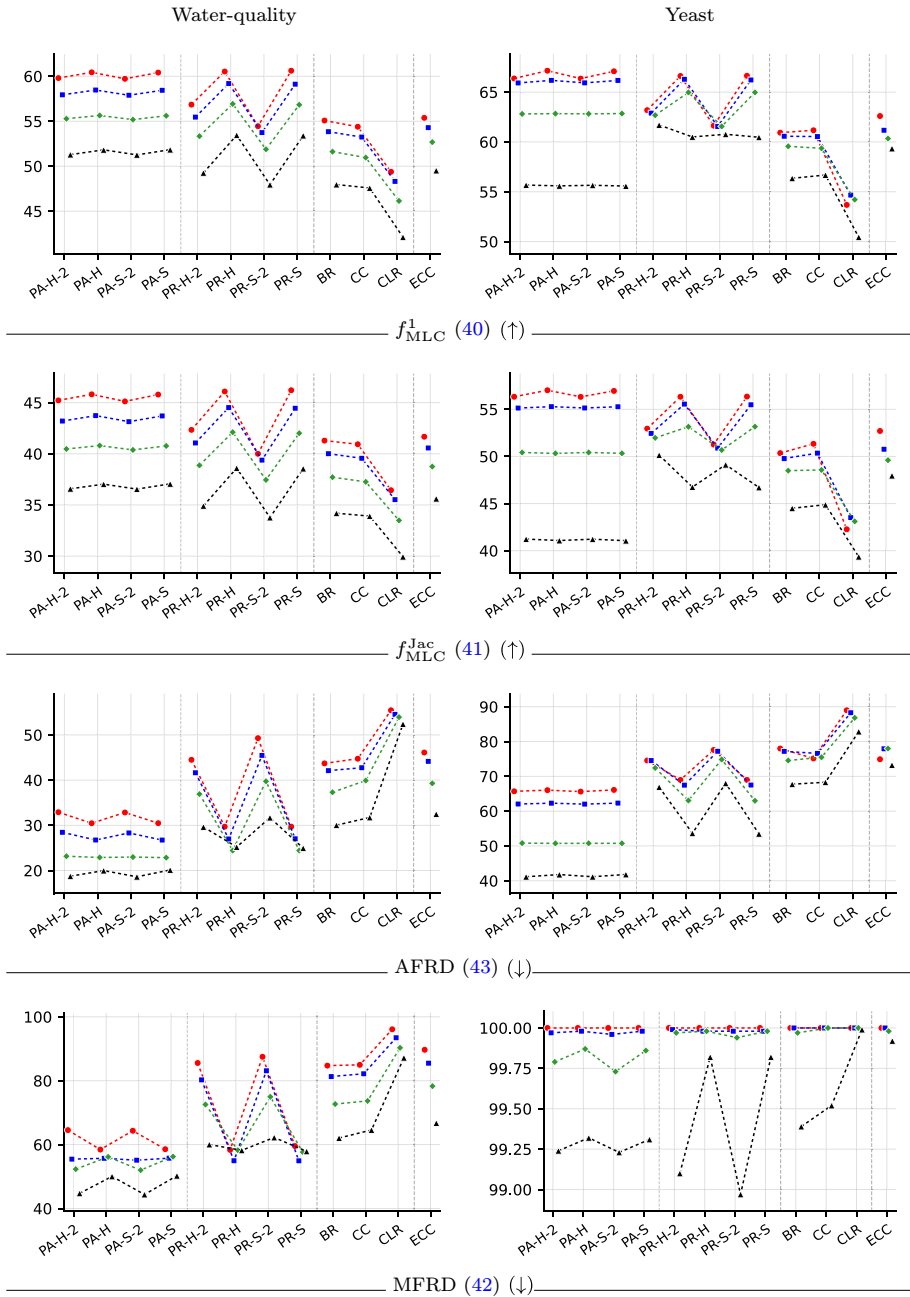


Table 17 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

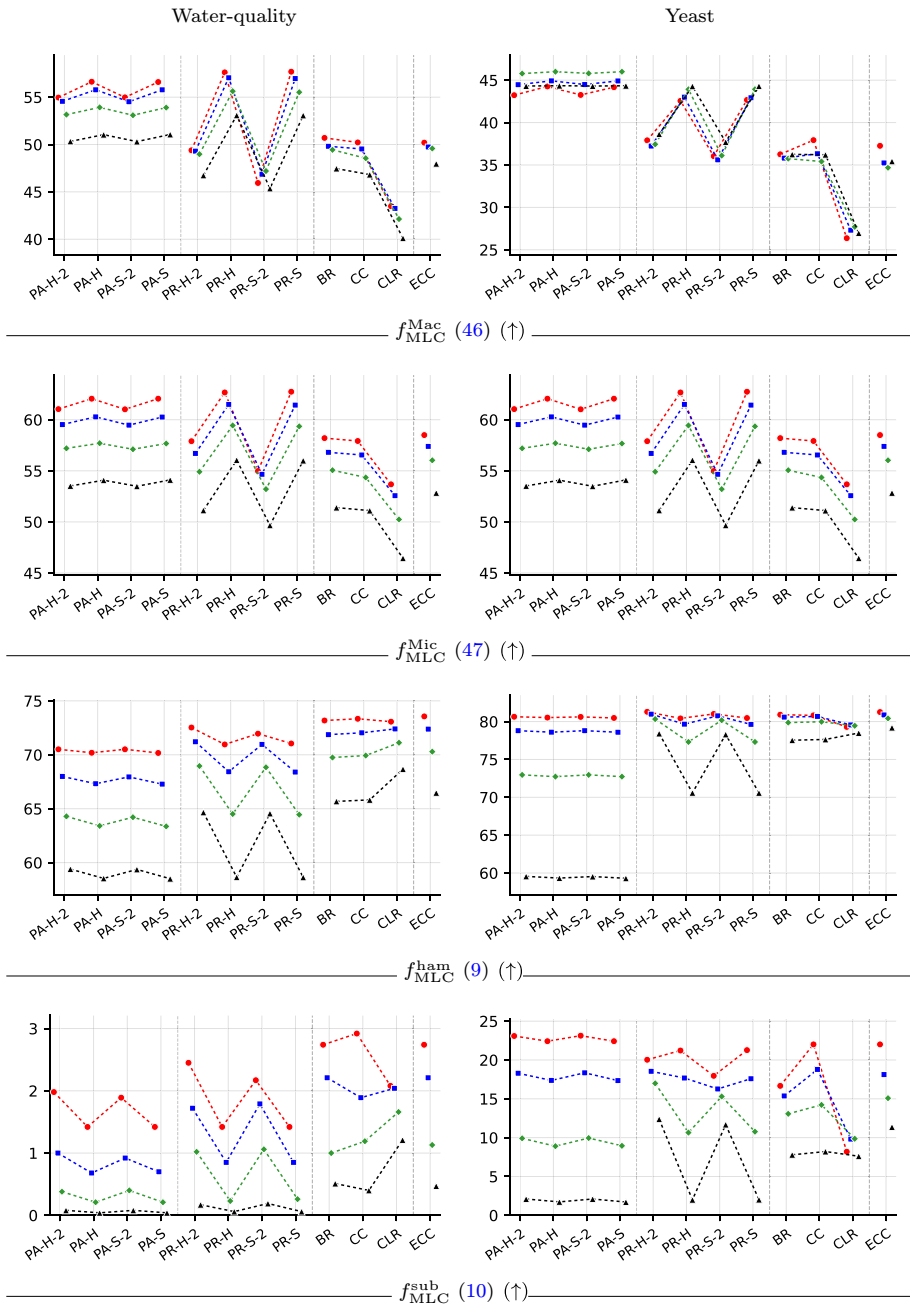


Table 18 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

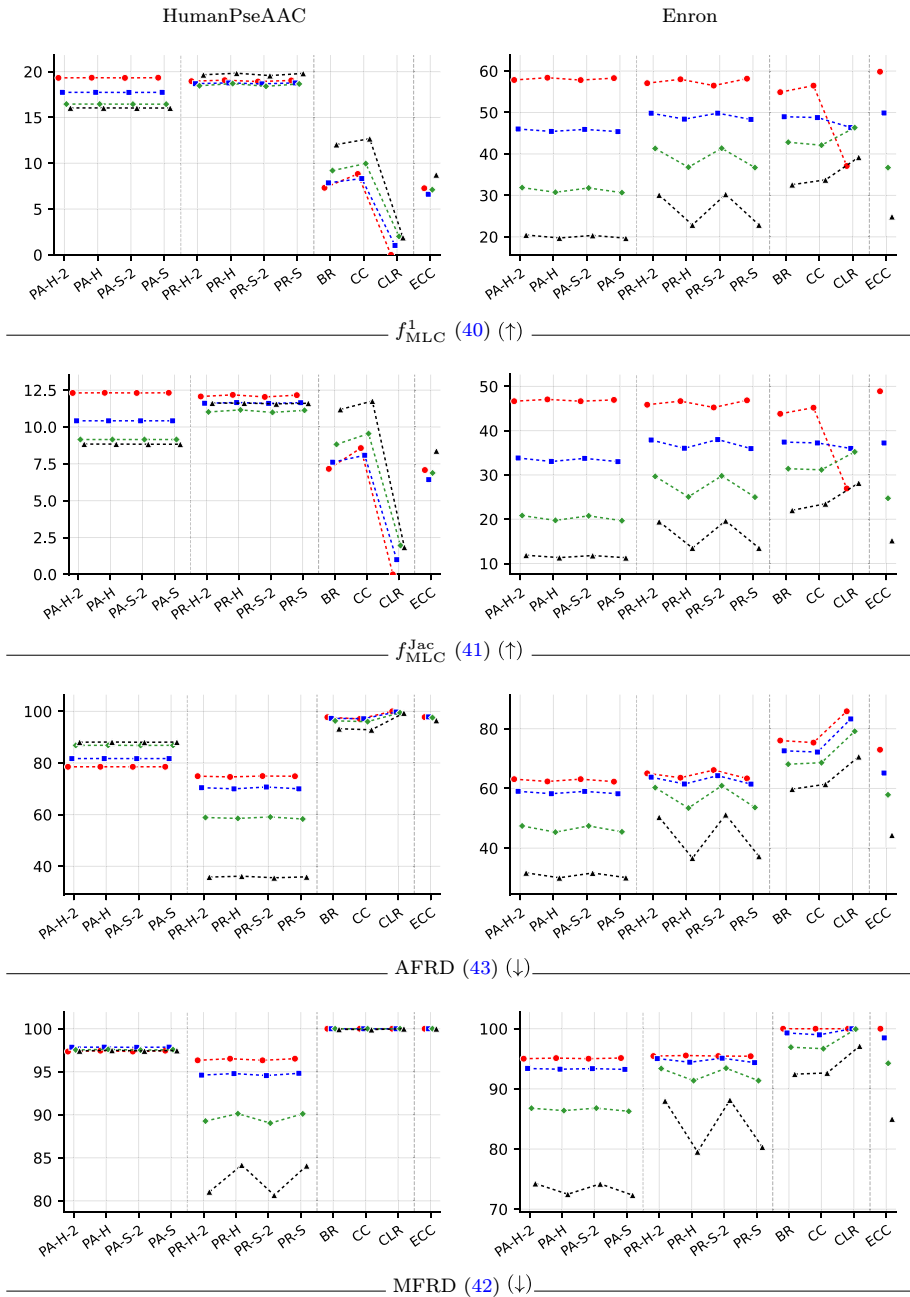
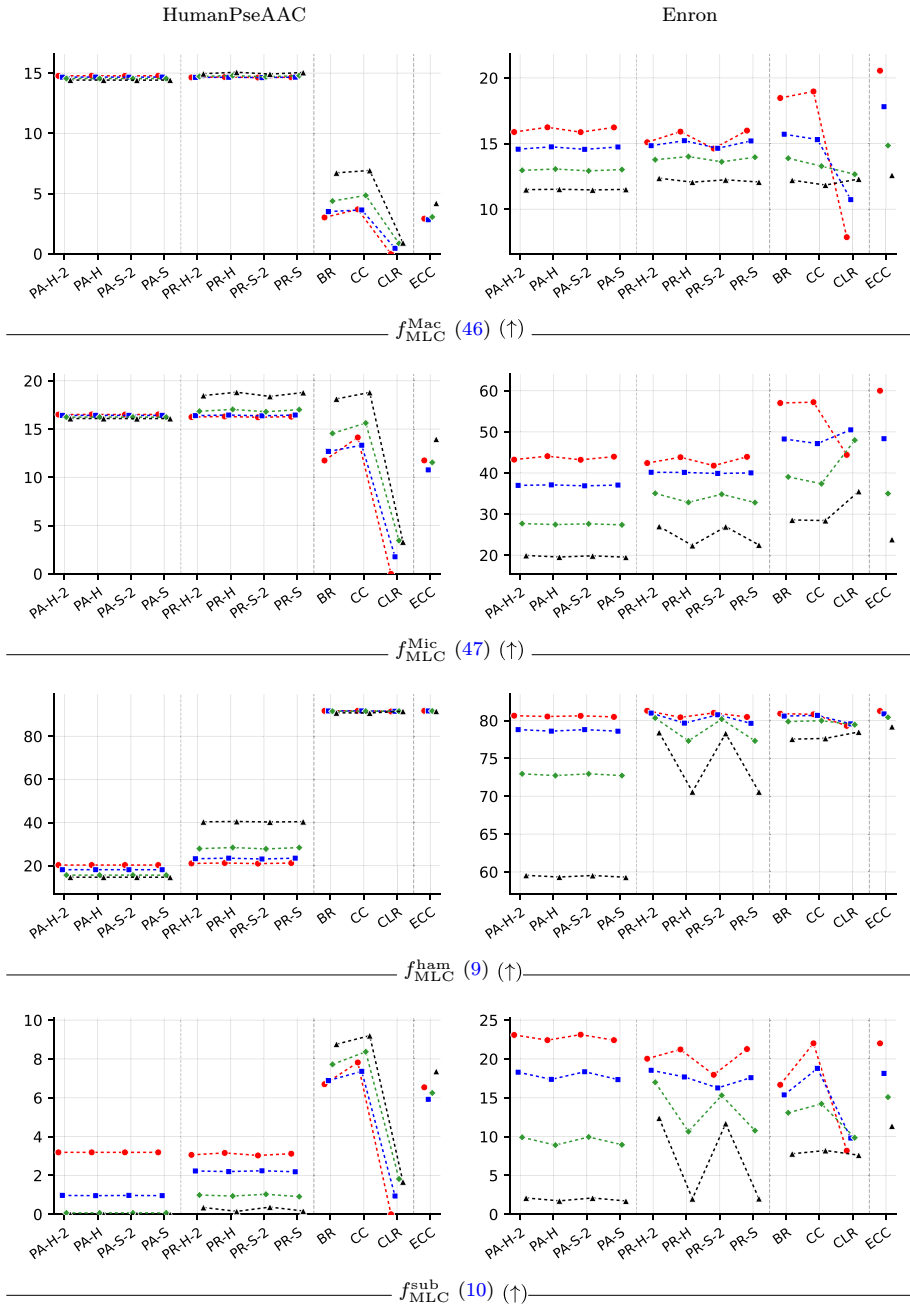


Table 19 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)



Predictions with Partial Abstention

A summary of the results for predictions with partial abstention is given in Table 20.

See Tables 21, 22, 23, 24, 25, 26, 27 and 28.

Table 20 Predictions with partial abstention: Names of datasets and the tables with corresponding results

Names	Place
GpositivePseAAC & PlantPseAAC	Tables 5 and 10
CHD_49 & Emotions	Tables 21 and 22
VirusPseAAC & Scene	Tables 23 and 24
Water-quality & Yeast	Tables 25 and 26
HumanPseAAC & Enron	Tables 27 and 28

Average Results

The average results across the data sets with random forest are given in Fig. 29. It confirms that classifiers opting for preorders, especially (PR-H, PR-S), consistently outperform (BR, CC, CLR, ECC) in terms of f_{MLC}^1 (40). Moreover, allowing predictions with abstention often leads to impressive improvement in terms of f_{MLC}^1 (40). The results on abstention rates, i.e., ABS (49) and AABS (48), confirm that classifiers opting for preorders can gain competitive improvement in terms of f_{MLC}^1 (40) with significantly lower abstention rates, compared to classifiers opting for partial orders, i.e., (PA-H-2, PA-H, PA-S-2, PA-S).

We will only provide the average results across the data sets with LightGBM as the base learner. The results given in Fig. 30 suggest a few consistent trends. In terms of f_{MLC}^1 (40), classifiers opting for preorders, i.e., (PR-H-2, PR-H, PR-S-2, PR-S), consistently outperform (BR, CLR). They are slightly better than CC on noisy data and are competitive with ECC. Allowing predictions with abstention often leads to promising improvement in terms of f_{MLC}^1 (40), but is less impressive compared to the ones achieved with random forests as the base learner. Finally, the results on abstention rates, i.e., ABS (49) and AABS (48), confirm that classifiers opting for preorders can gain competitive improvement in terms of f_{MLC}^1 (40) with significantly lower abstention rates, compared to classifiers opting for partial orders.

For both base learners, a similar trend, but a bit less clear, is also observed for f_{MLC}^{mac} (46), which is another balancing metric. The results are omitted to save space.

Table 21 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

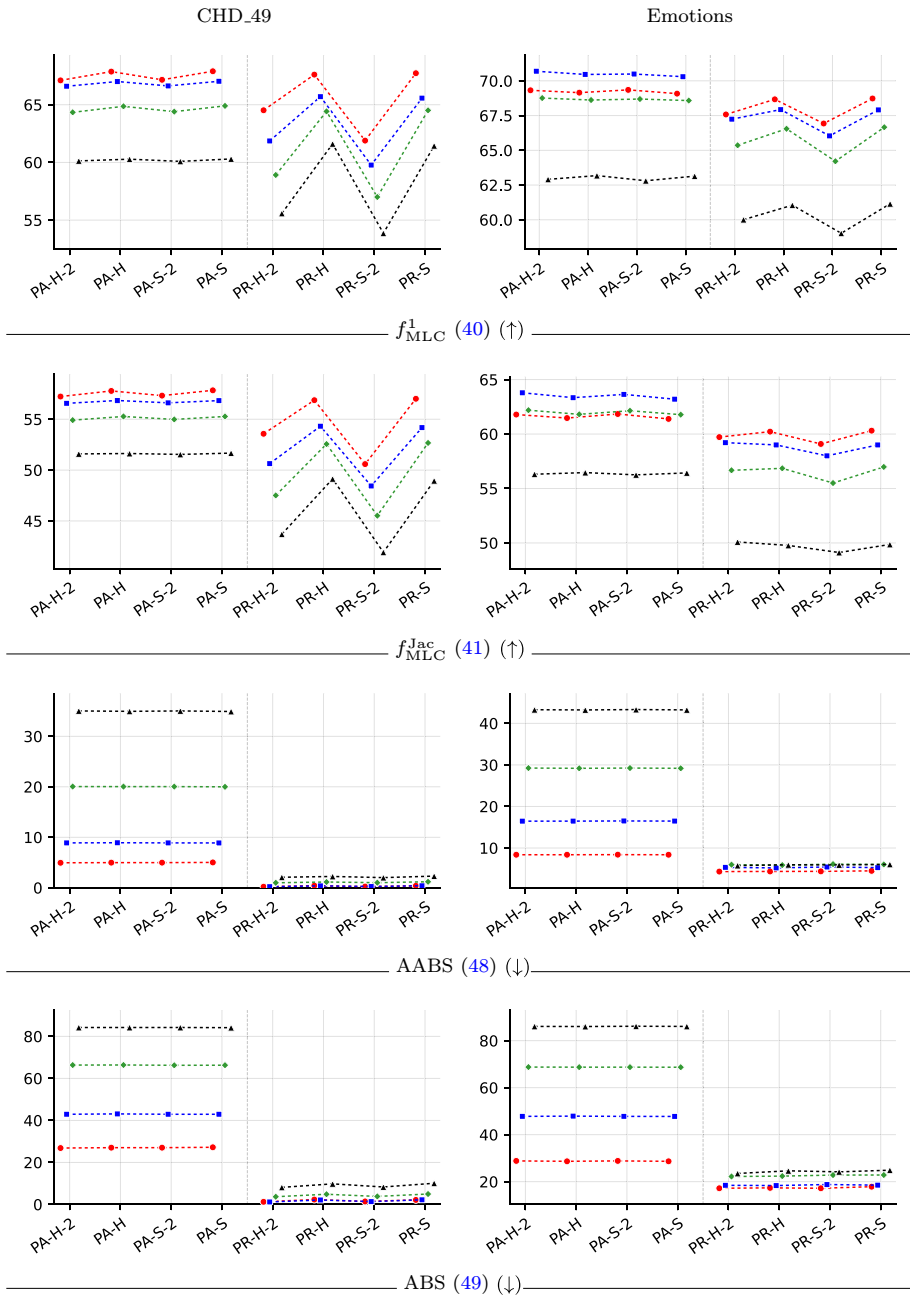


Table 22 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

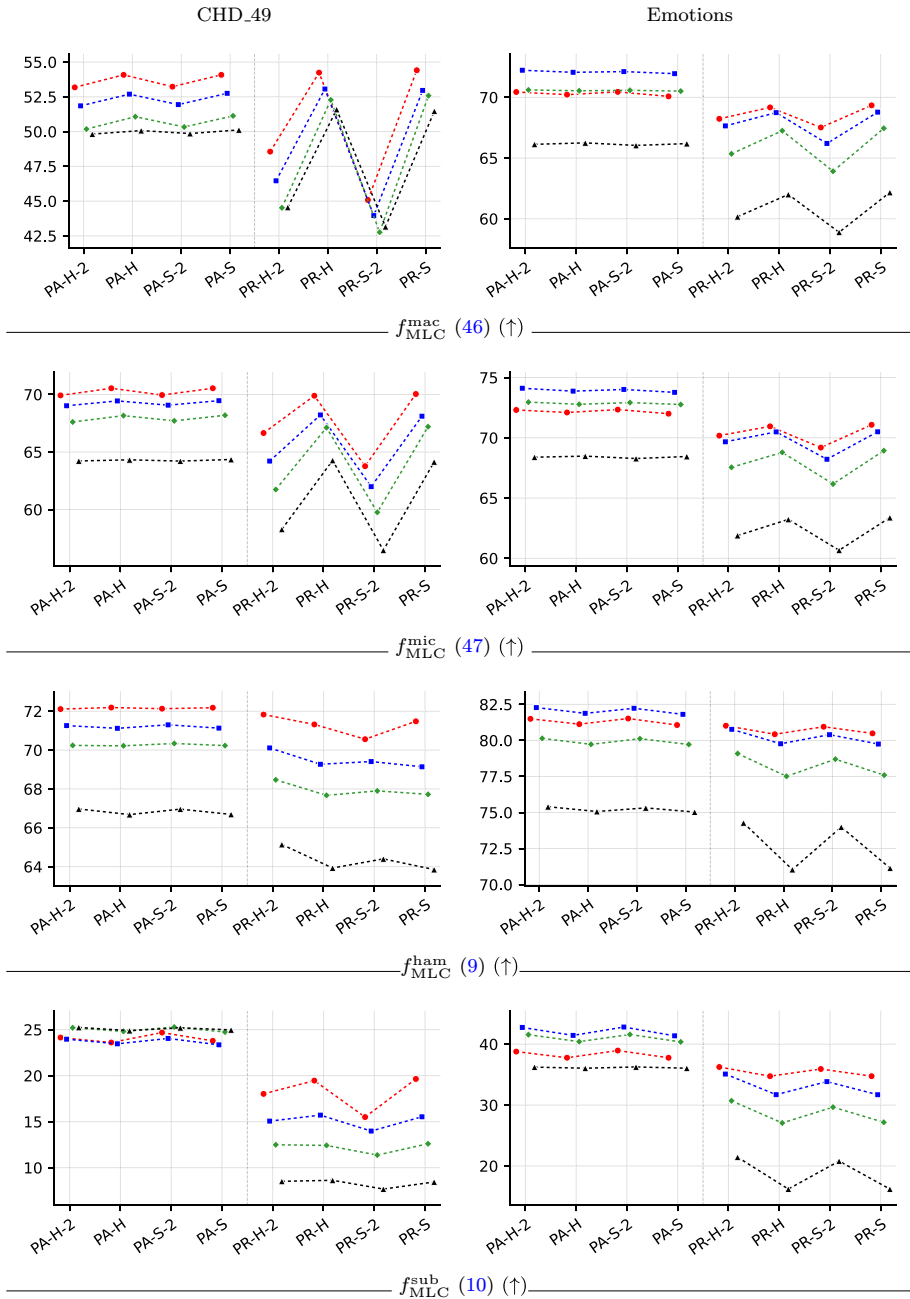


Table 23 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

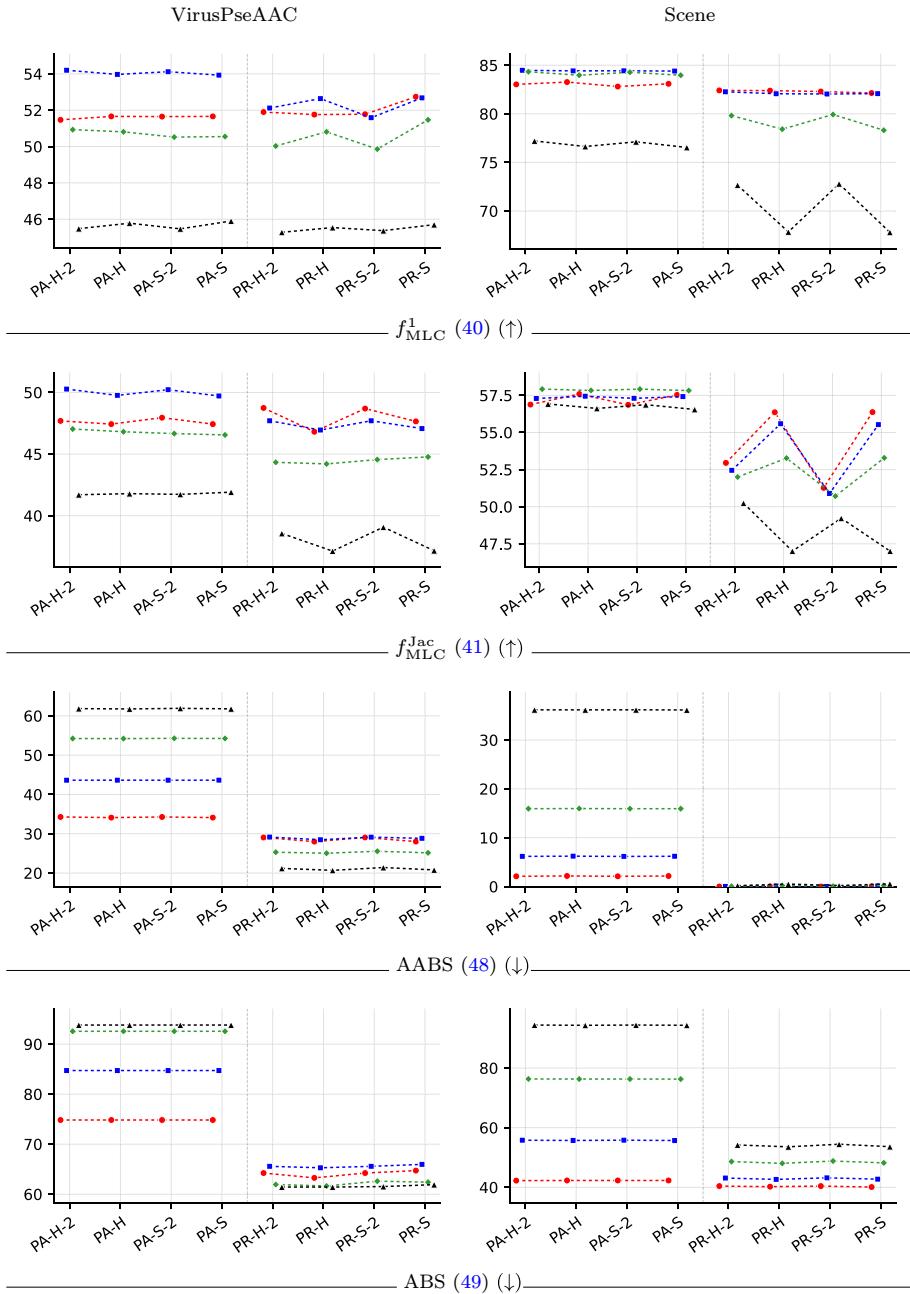


Table 24 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

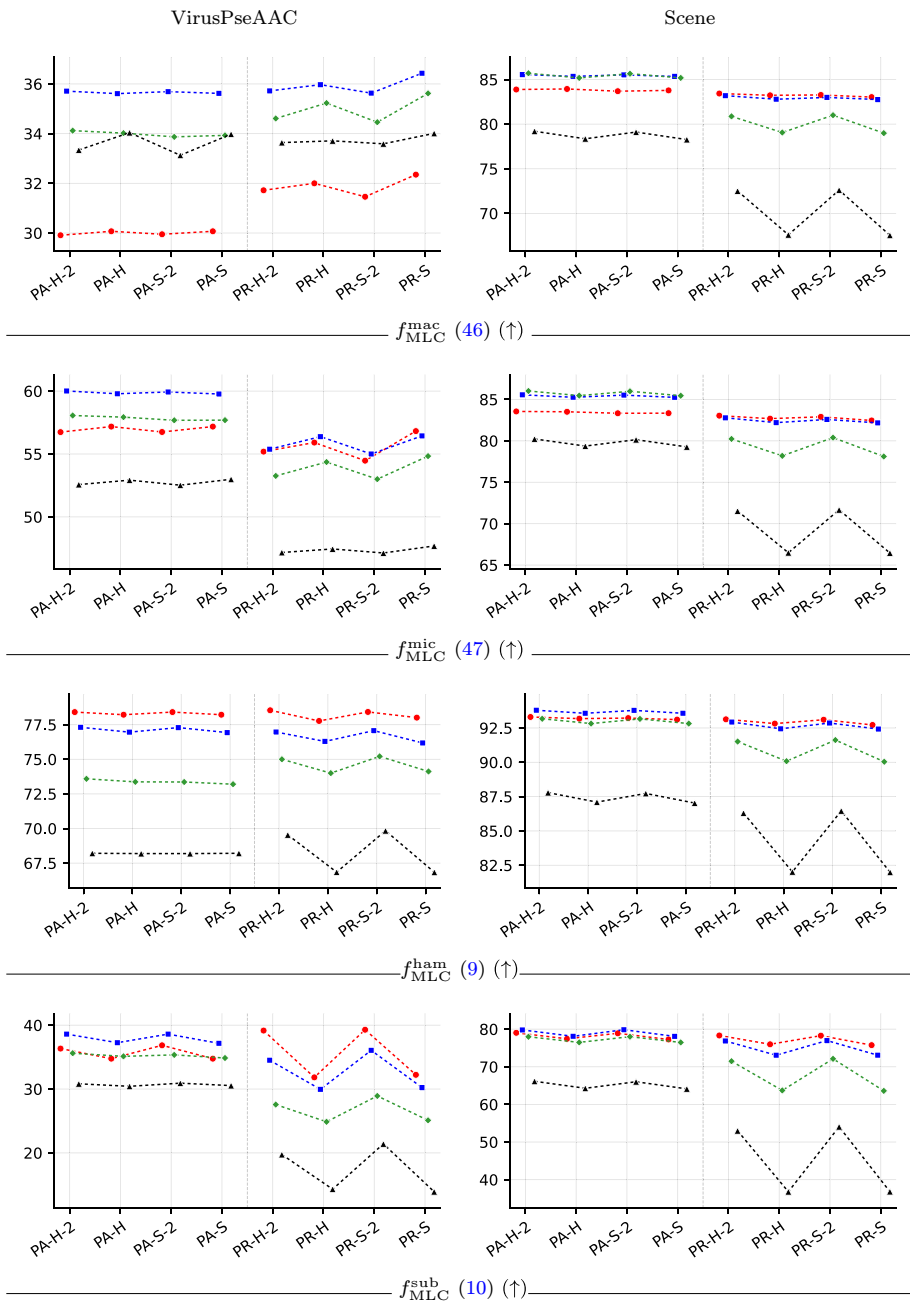


Table 25 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

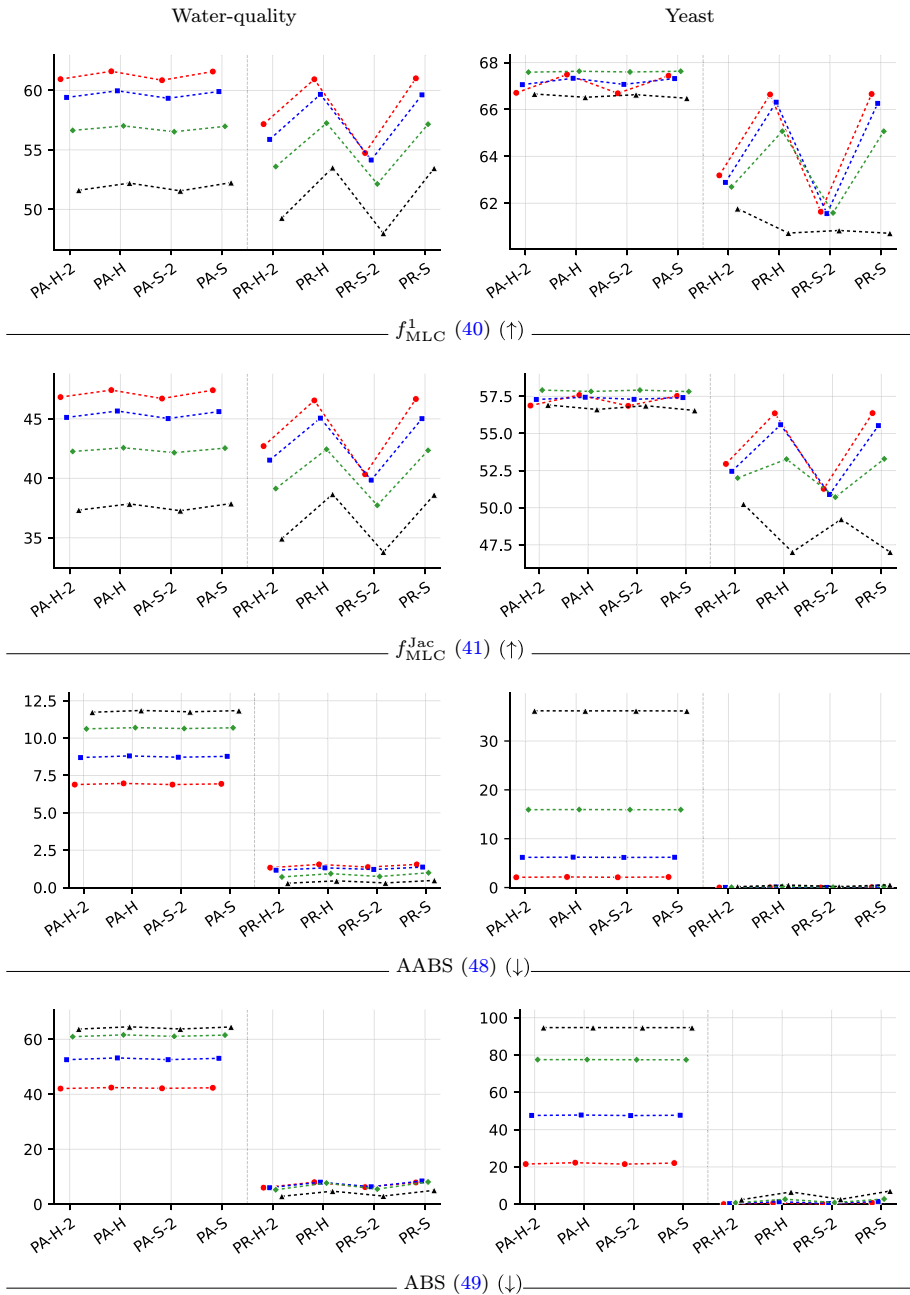


Table 26 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

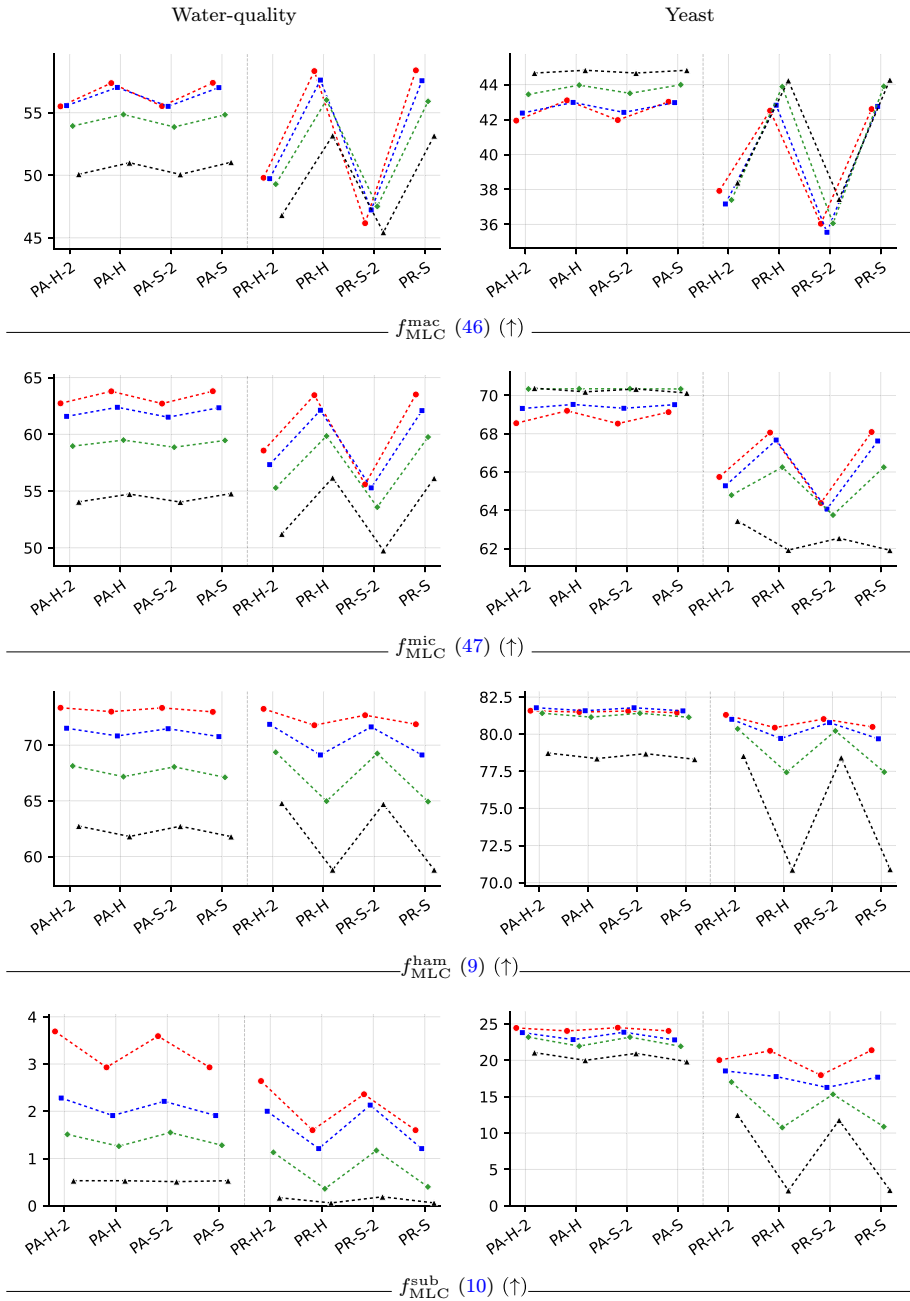


Table 27 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

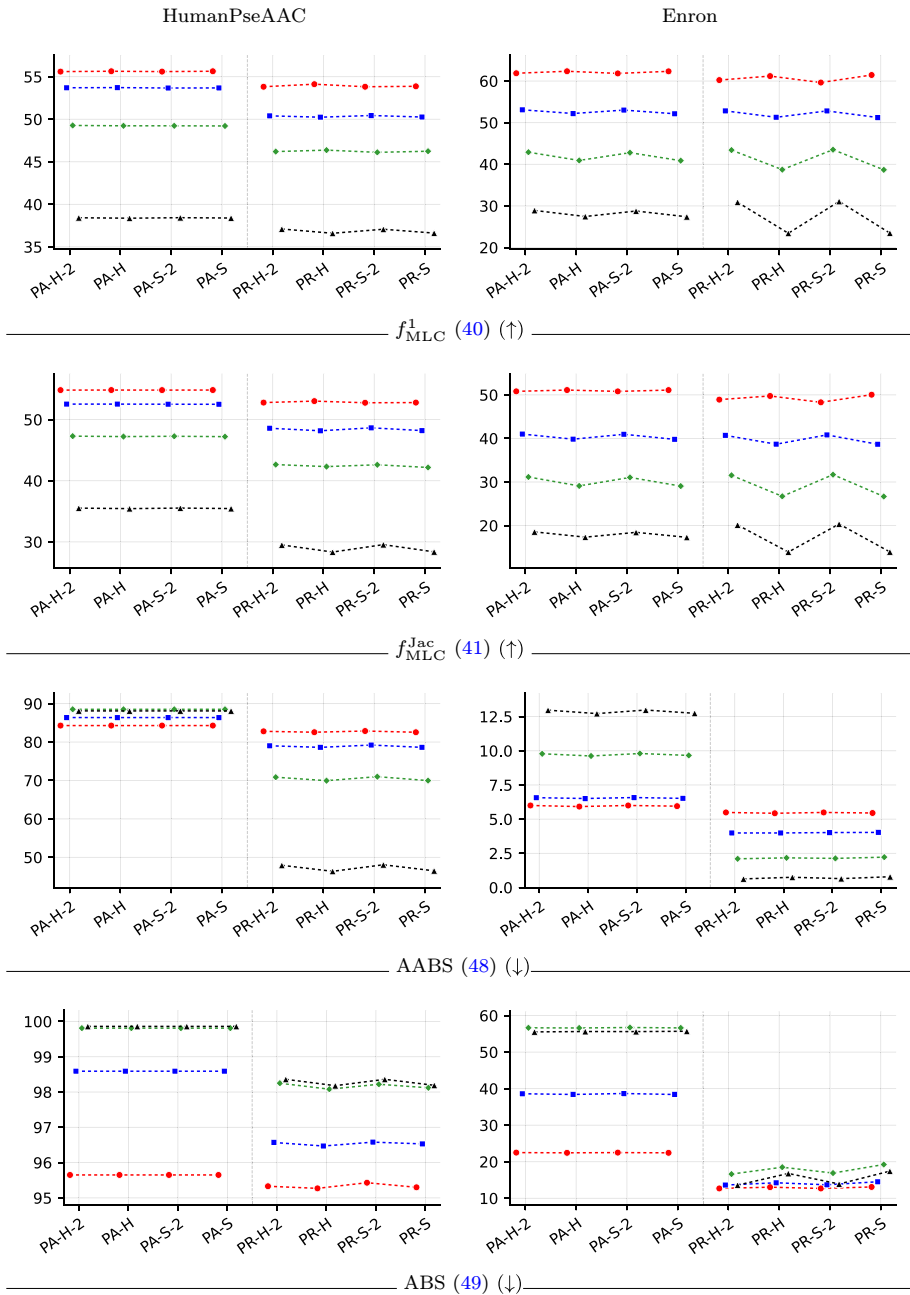


Table 28 Average scores (in %, y-axis) over 5×5 cross-validation folds. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$ (circle), $\alpha = 0.1$ (square), $\alpha = 0.2$ (losange) and $\alpha = 0.3$ (triangle)

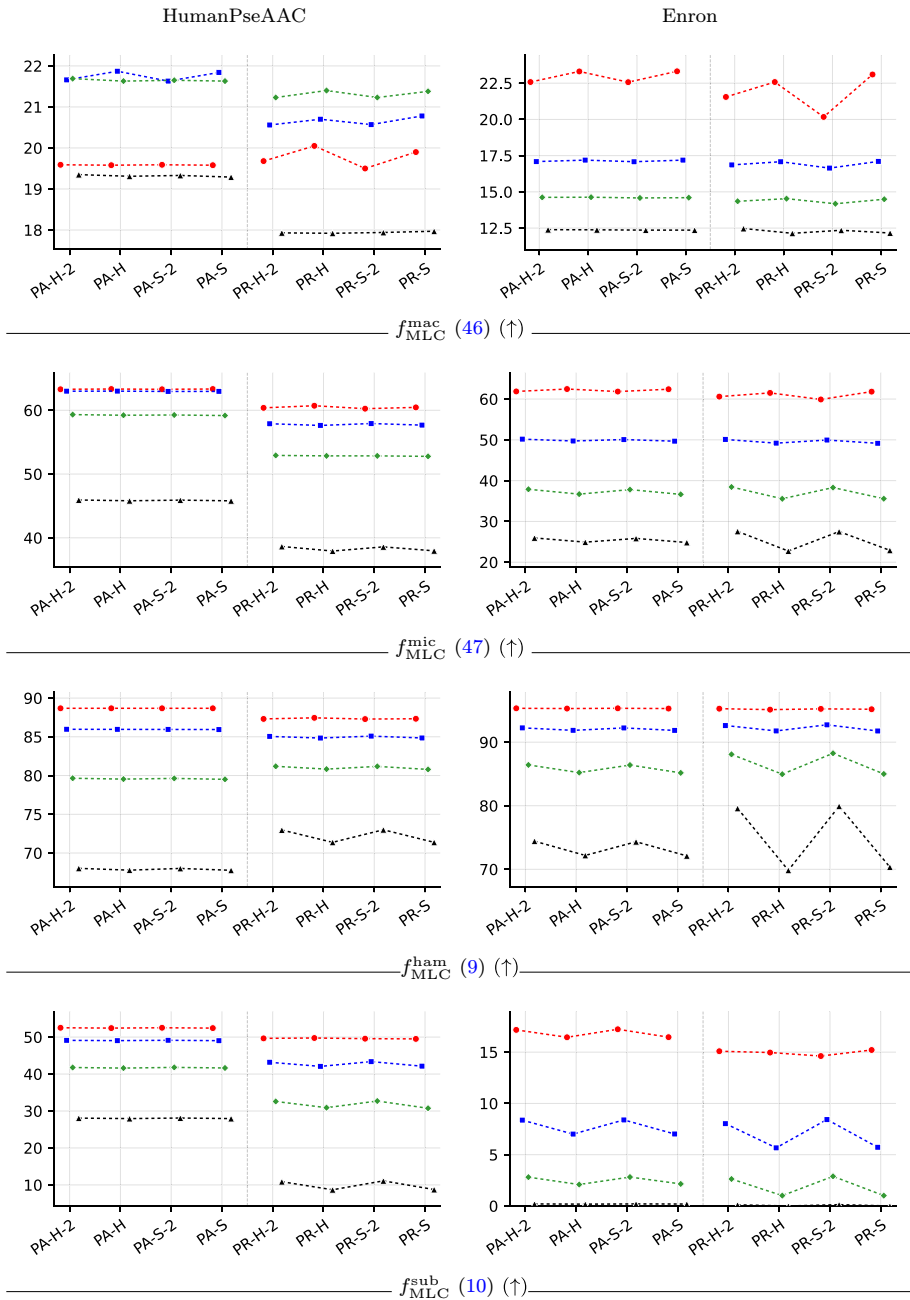


Table 29 Average scores (in %, y-axis) across the datasets with random forests. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$. In the right column, ABS (49) and AABS (48) are encoded as the bars and the white dots, respectively

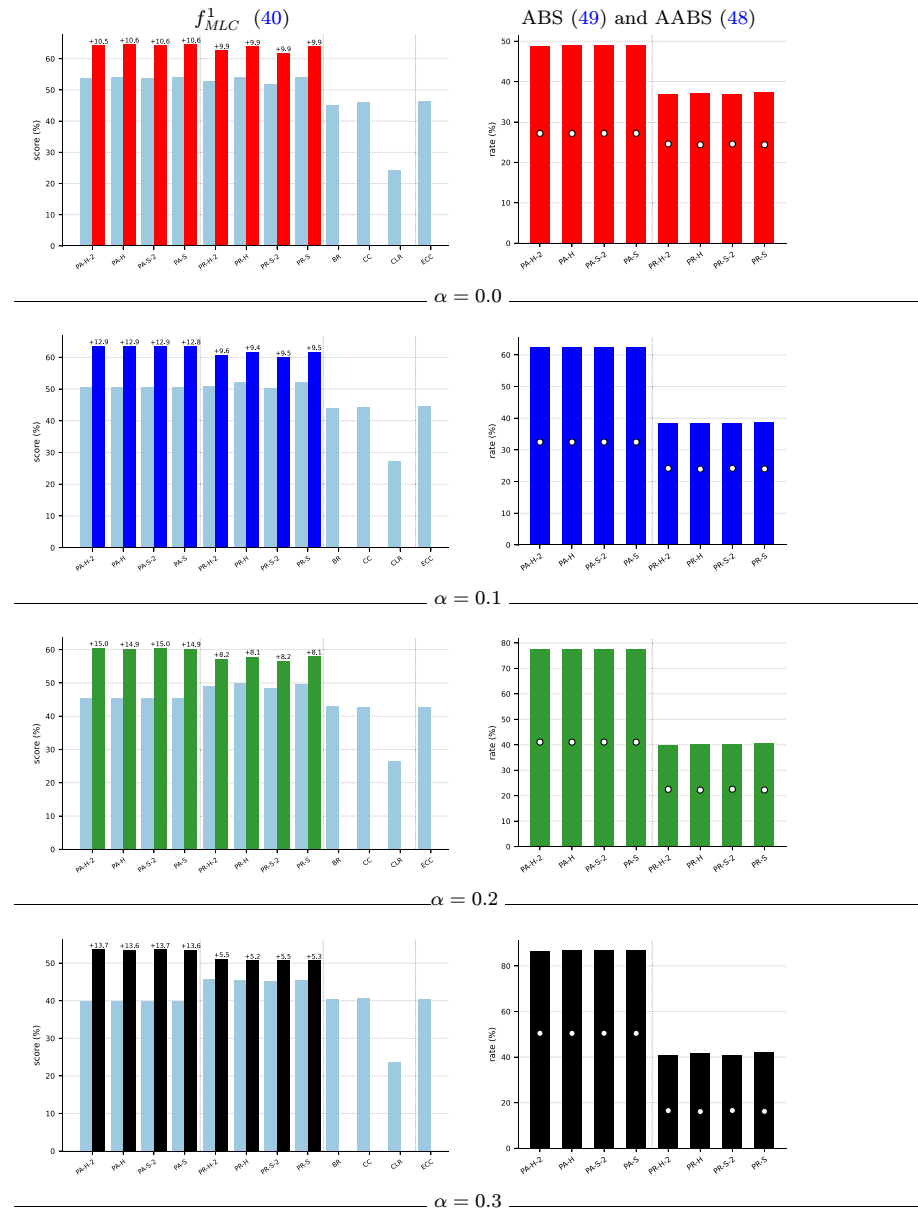
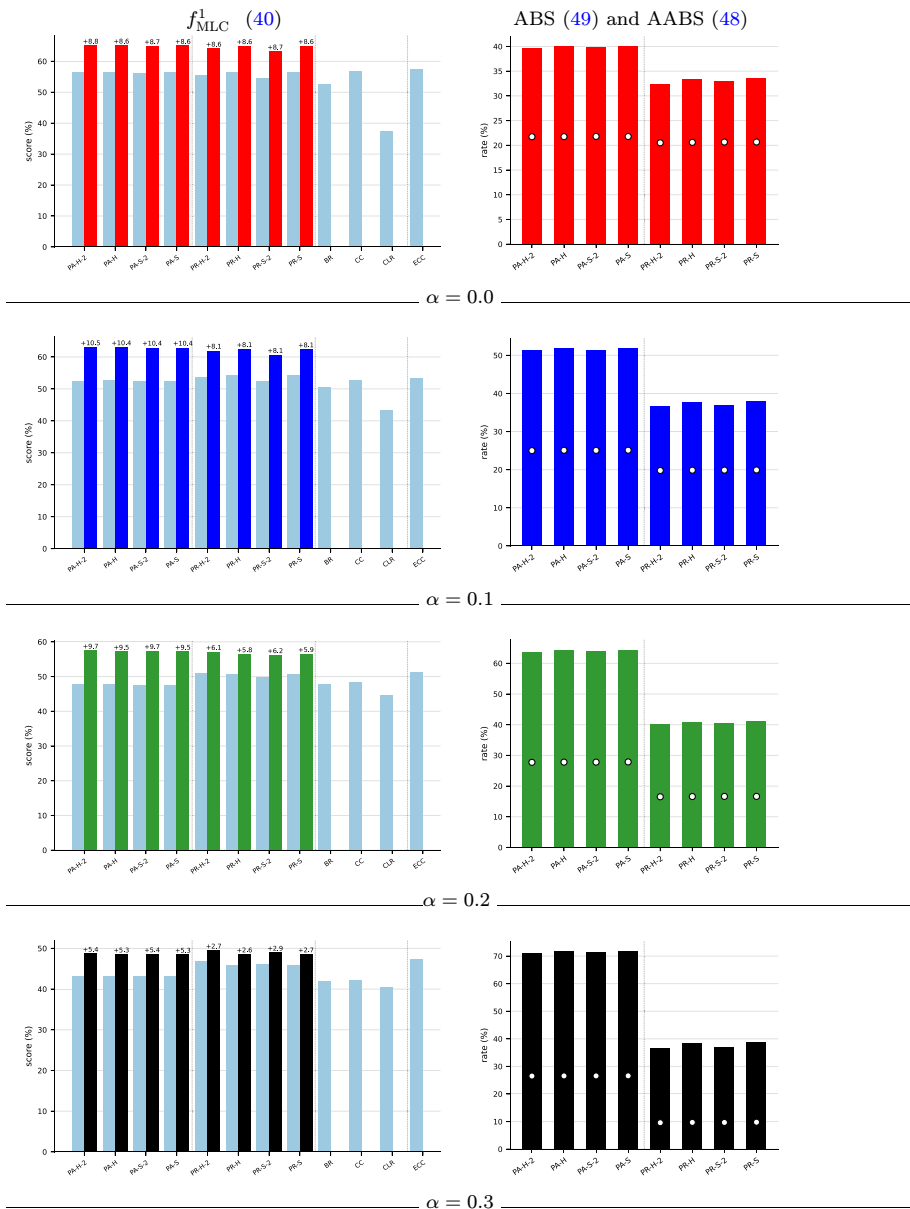


Table 30 Average scores (in %, y-axis) across the datasets with LightGBM. Results are color-coded w.r.t. the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$. In the right column, ABS (49) and AABS (48) are encoded as the bars and the white dots, respectively



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Data Availability The 10 data sets used in the empirical study are all publicly available at <https://cometa.uja.es/datasets/>. The source code has been made available at <https://github.com/hxtruong6/pre-order-for-mlc>.

Declarations

Conflict of interest We declare that we have no conflict of interest.

Ethical Approval We declare that this research did not require Ethics approval.

Consent for Publication All the authors of this manuscript consent to its publication.

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